

Flashes in the Dark: New Searches for Axions and Macroscopic Dark Matter

Kevin Zhou



Stanford
University



Johns Hopkins Seminar— September 7, 2022

The Best of Times



LHC Stakes *1.49 million furlongs per second.*
Purse: Nobel Prize for long-sought particles, well-motivated theoretical concepts, and speculative guesses.



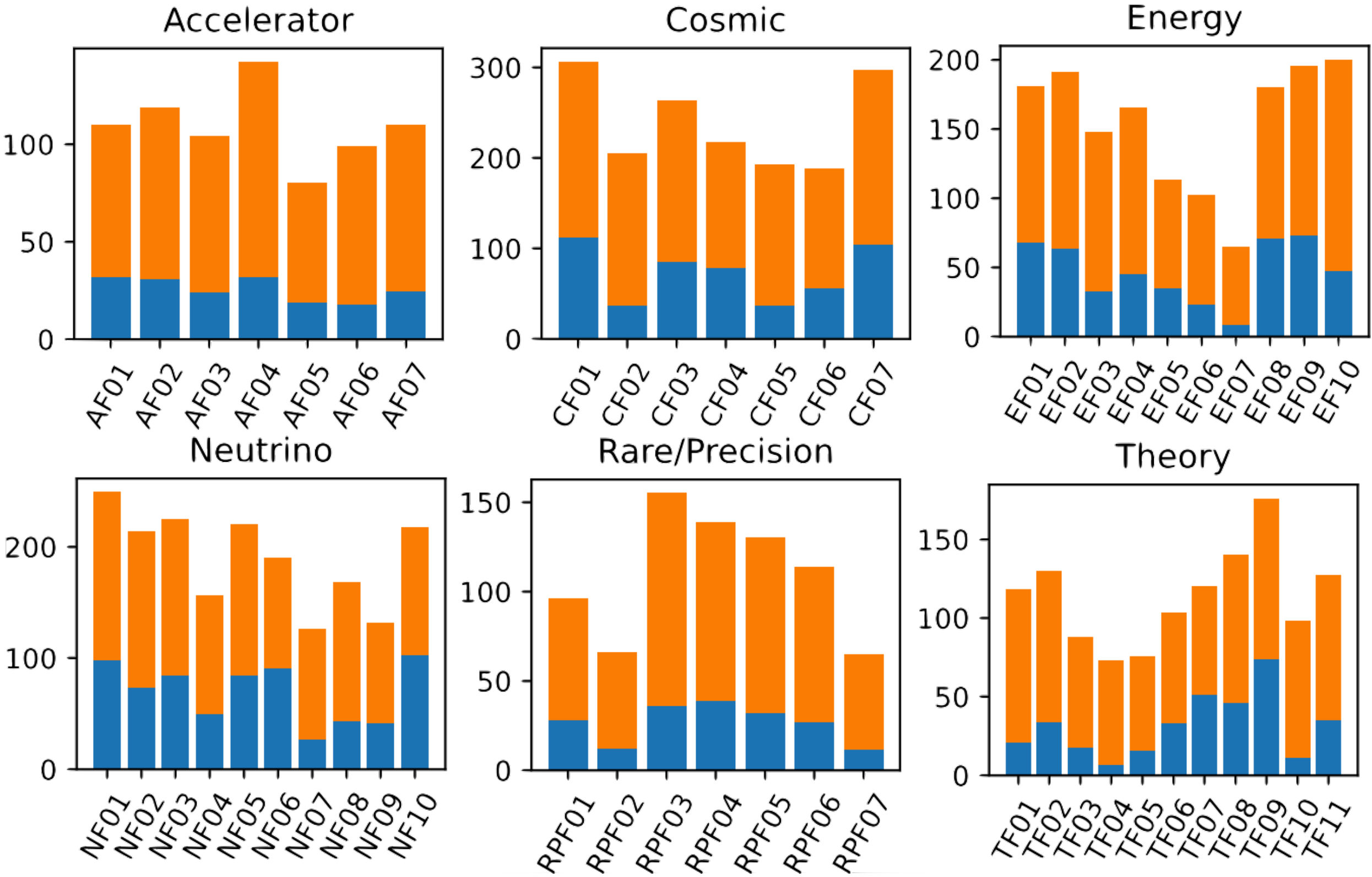
DISCOVERY	COMMENTS	ODDS*
Standard Model Higgs	As good as discovered, some say	2–1
Big surprises	Expect the unexpected	2–1
Supersymmetry	Too beautiful to be wrong?	5–1
Extra dimensions	More an inspired guess than a prediction	14–1
Composite electrons, etc.	Bigger fleas have littler fleas ...	14–1
Leptoquarks	Weird particles hinted at by another collider	49–1
Nothing	Be careful what you wish for	7–1

* Based on survey of roughly 300 grad students, postdocs, and professors at Fermilab, as reported in the January/February issue of *Symmetry Magazine*. Respondents could choose more than one prediction.

The New Frontier

Physics and Technology	Facilities	Colliders		
		Lepton Lepton	Lepton Hadron	Hadron Hadron
Testing the Standard Model Electroweak QCD etc.				
Beyond the Standard Model Technicolor Supersymmetry Grand Unification etc.				
Limits of Accelerator Technology and Novel Accelerator Ideas				
Novel Detector Ideas				

Snowmass 1982 Summary

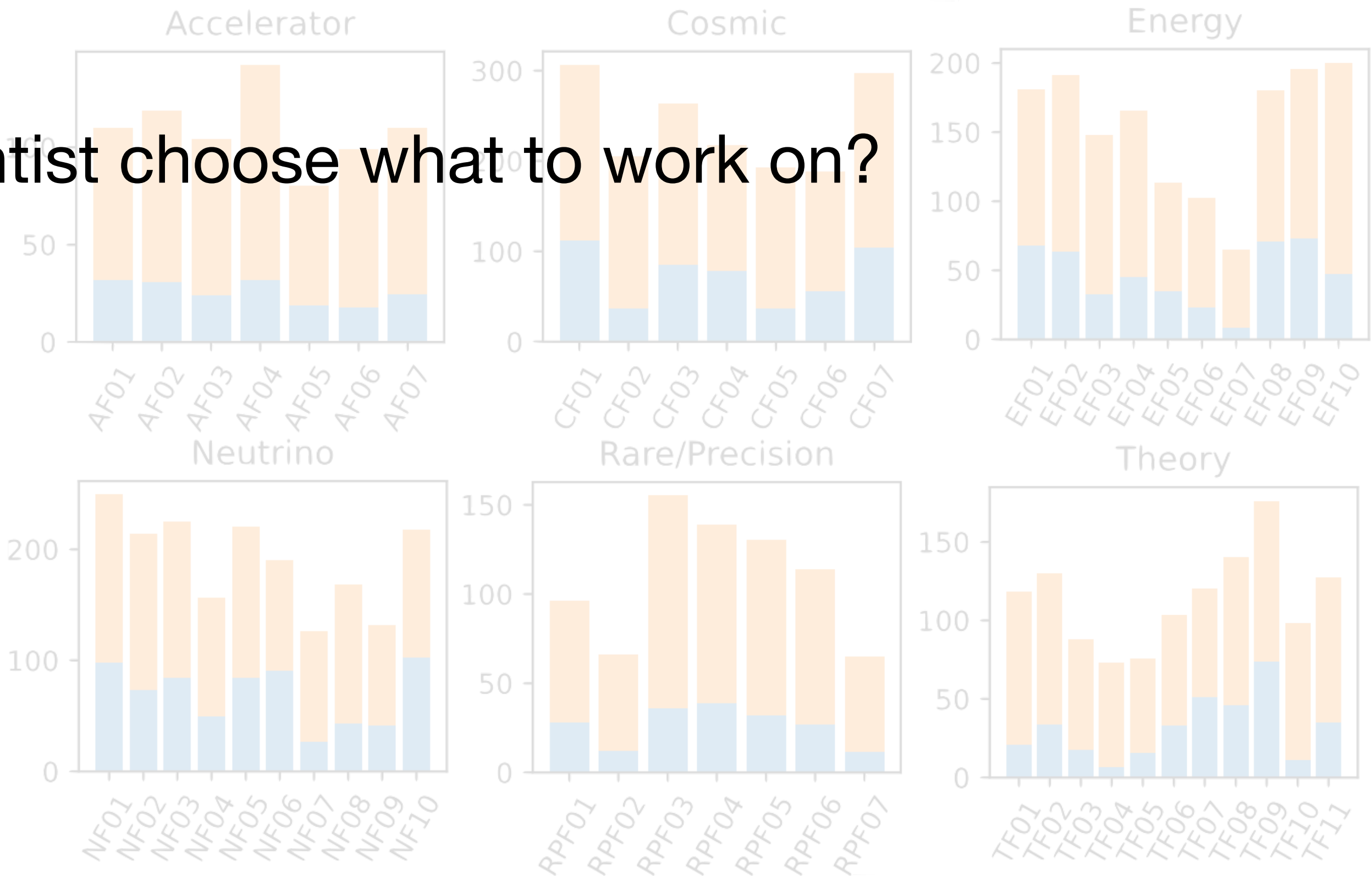


Snowmass 2021 Survey (2203.07328)

The New Frontier

How does a young scientist choose what to work on?

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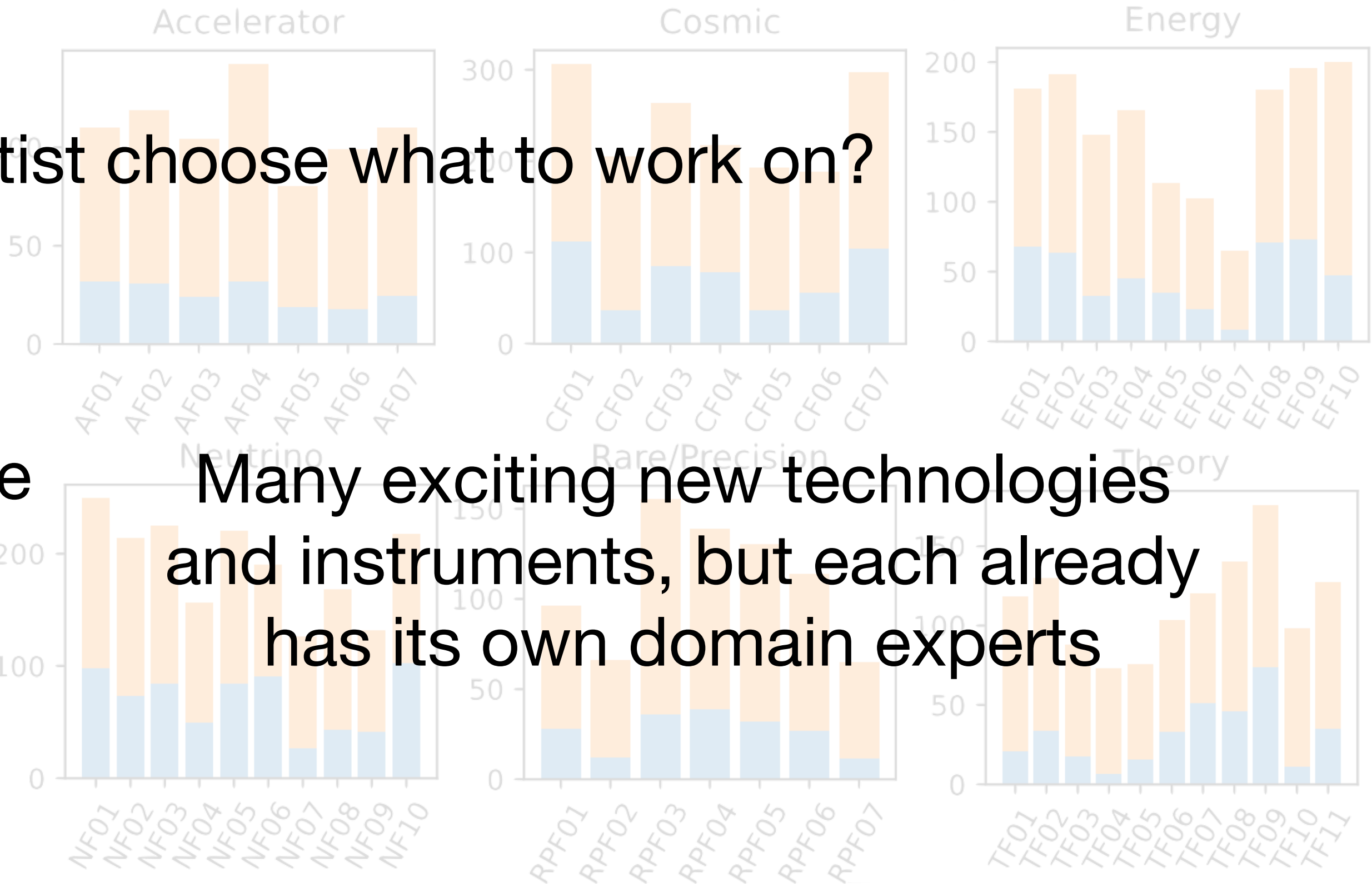
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How does a young scientist choose what to work on?

Many interesting BSM theories, but none has anywhere near the odds that supersymmetry did 15 years ago!

Many exciting new technologies and instruments, but each already has its own domain experts



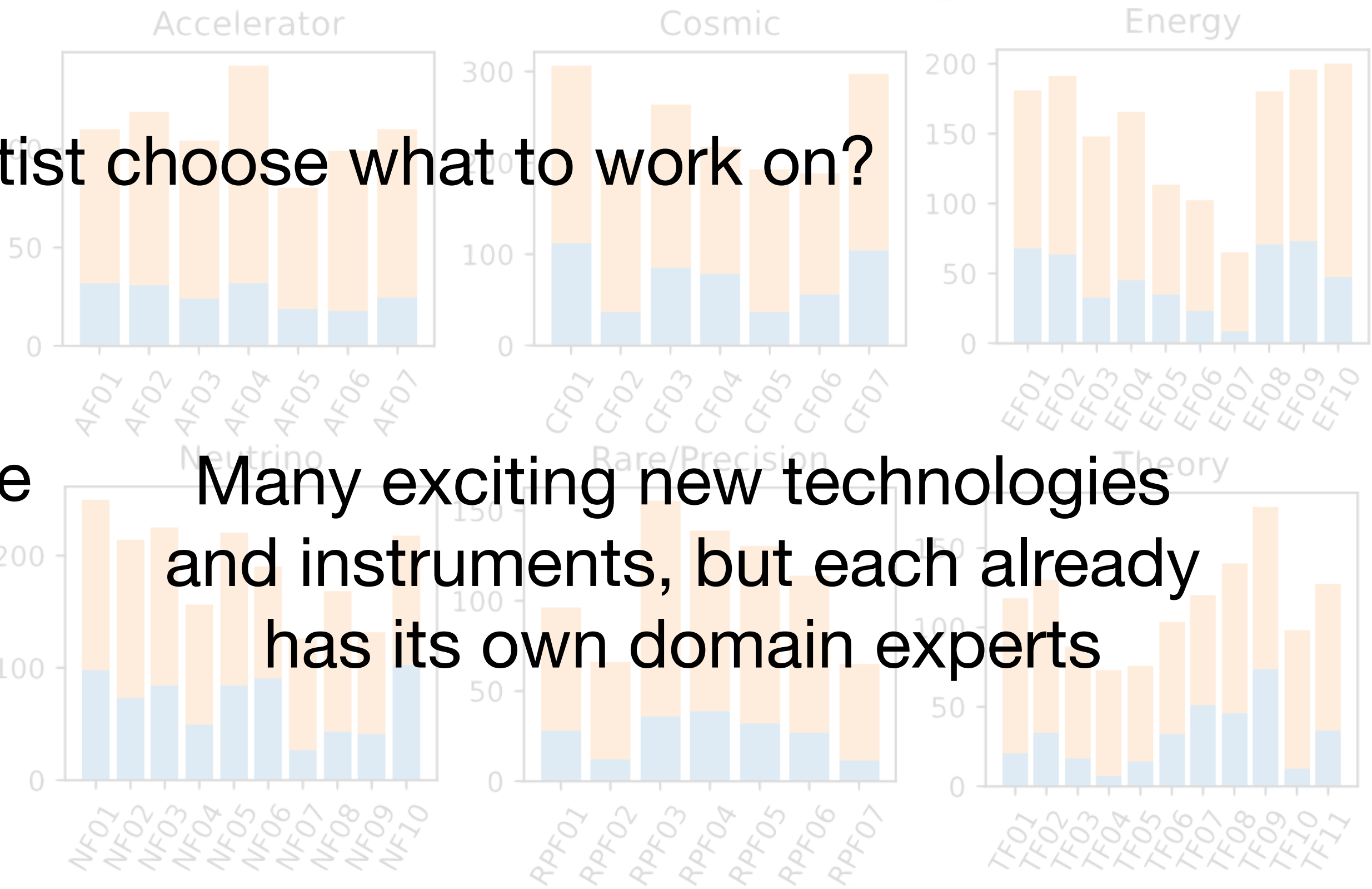
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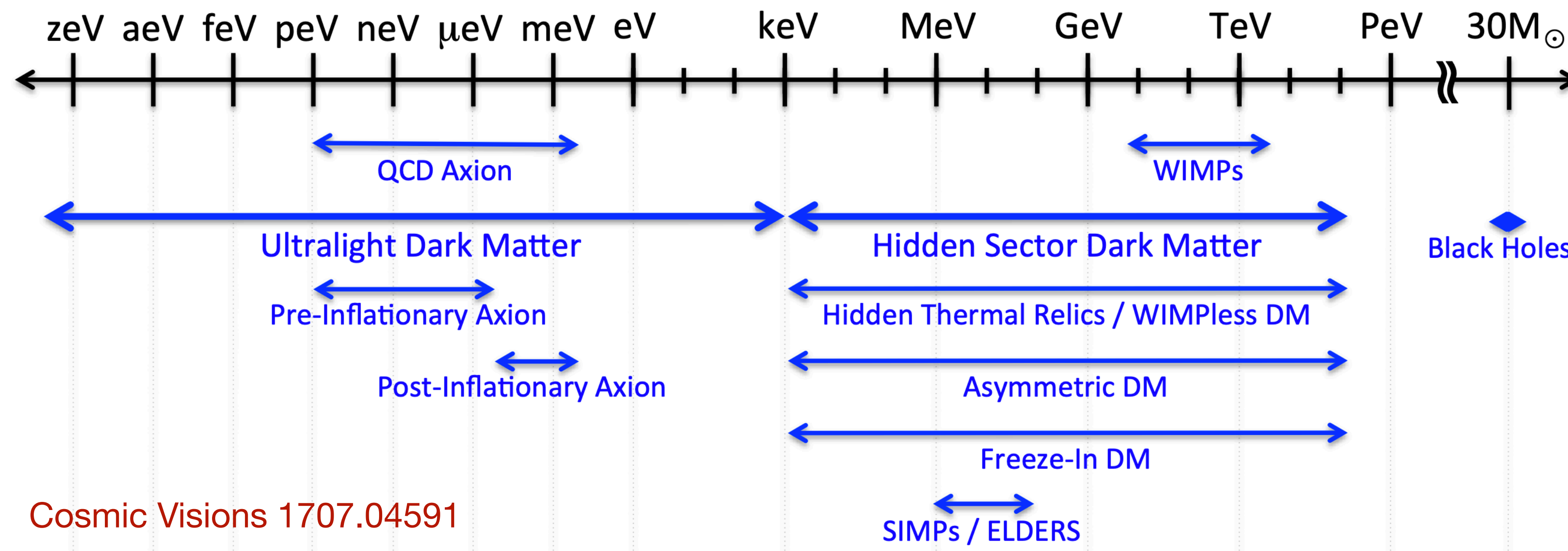
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Another path: linking theories and the technologies that can test them, across fields

Physics is a big place — many exciting connections have not been made!

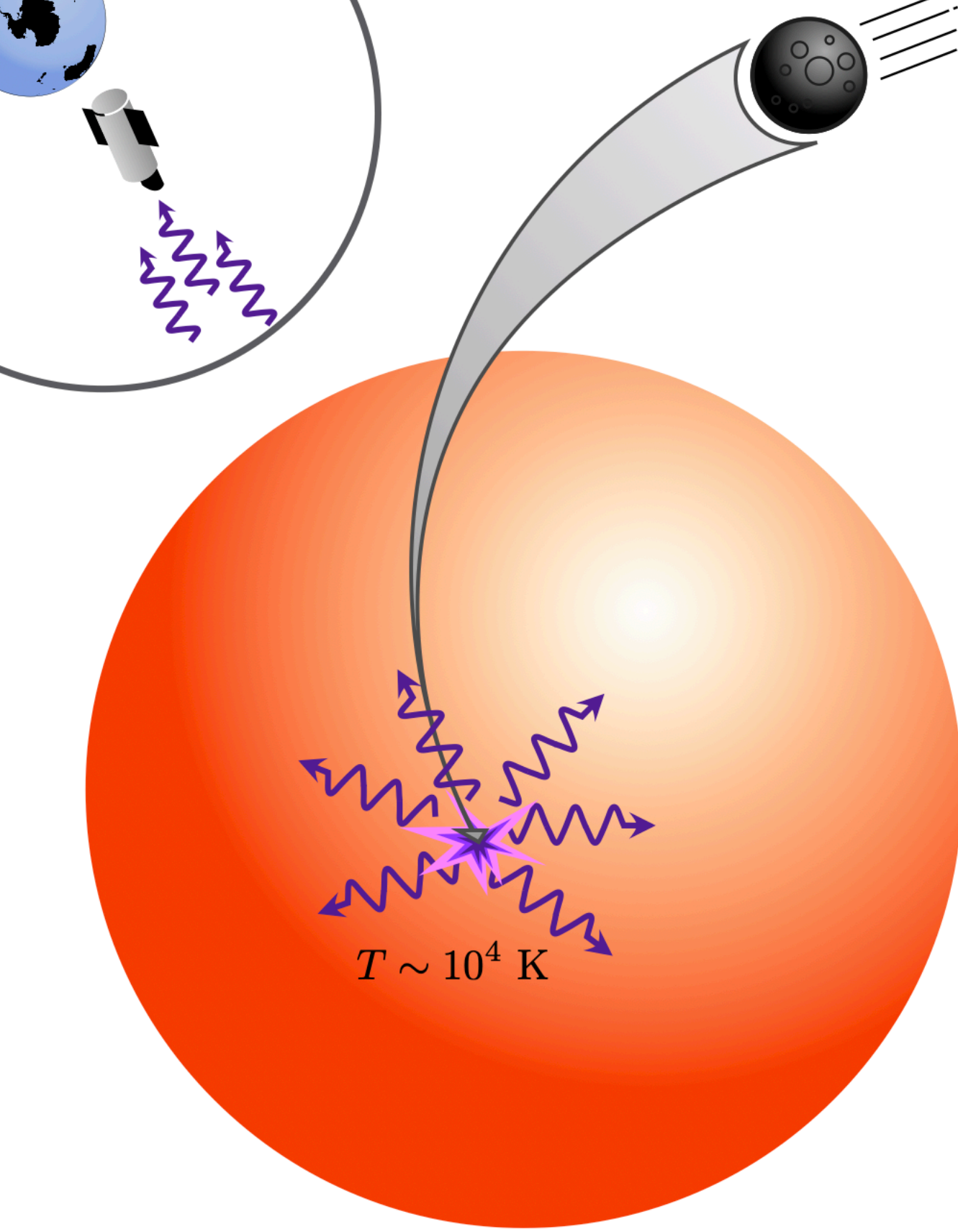
Case Study: Dark Matter



In current situation, we would like to both decisively test canonical models, and probe broadly into underexplored classes of models

New experimental programs can help do both!

Outline

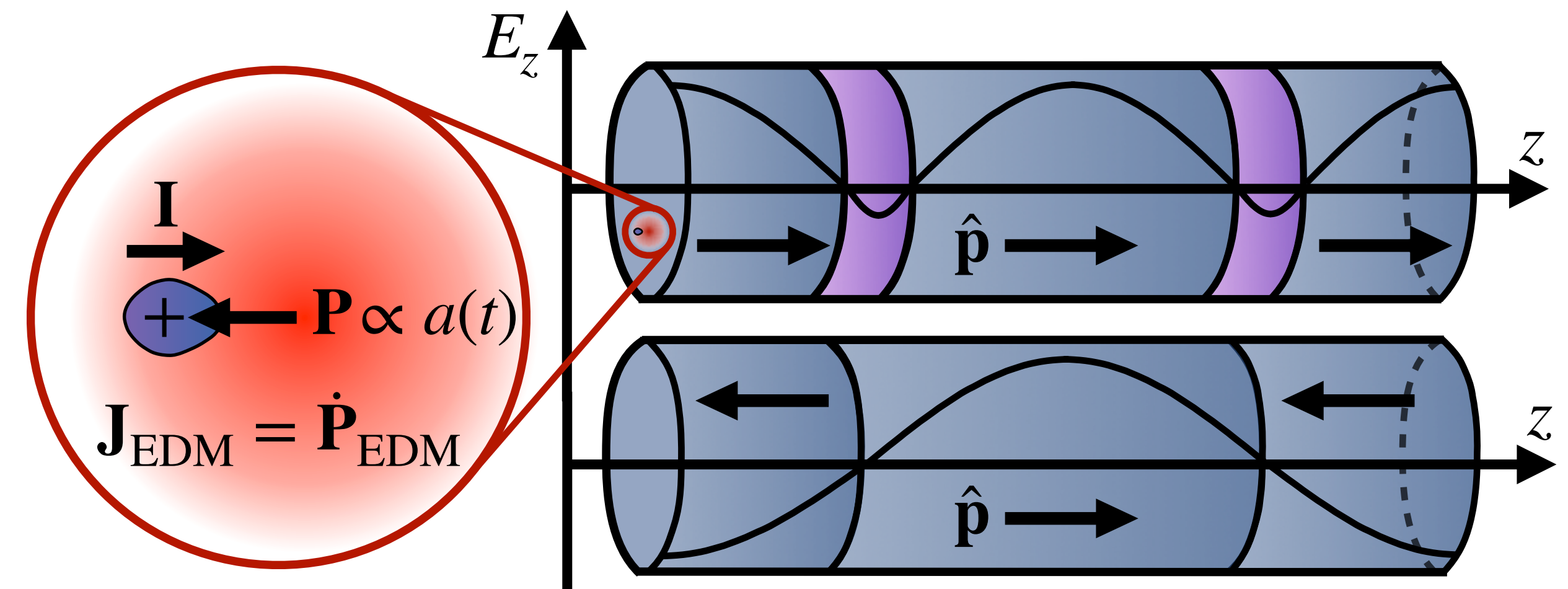


Searching for asteroid-mass macroscopic dark matter with ultraviolet stellar transients

Das, Ellis, Schuster, KZ (PRL 2022)

The first search for the QCD axion's defining coupling at GHz frequencies

Berlin, KZ (in prep)



Macroscopic DM

All direct evidence for dark matter is compatible with it being macroscopic:

$$M_{\text{pl}} = 10^{-39} M_{\odot} \quad \longleftrightarrow \quad 10^2 M_{\odot} \quad M_{\text{DM}}$$

- Must be composite above M_{pl}
- Region above $10^2 M_{\odot}$ strongly disfavored by dynamics (binary disruption, disk heating, evolution of dwarf galaxies and globular clusters)

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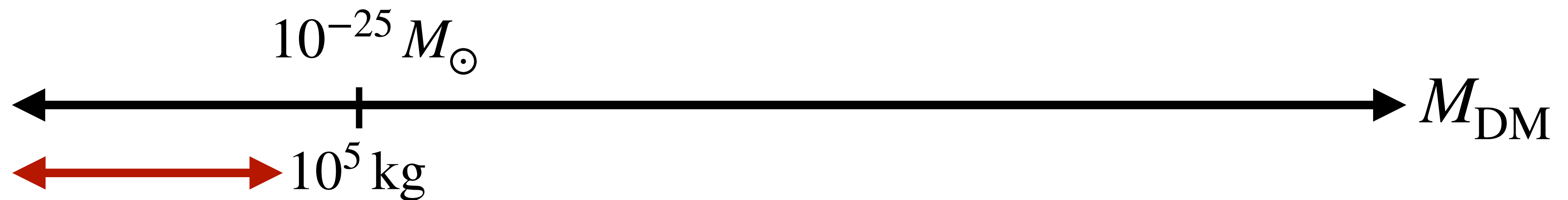
- Could be primordial black holes, but could also interact strongly with SM
- Parametrize by cross section σ for baryon scattering, where generically $\sigma = \pi R_{\text{DM}}^2$ for strong interactions without new long-range forces

Forming Macroscopic DM

- Several generic formation mechanisms which appear in many models
 - Direct fusion in asymmetric DM: “dark nucleosynthesis”
1411.3739, 1707.02316
 - Collapse through cooling via dark $U(1)$
1707.03829, 1812.07000
 - Phase transitions forming “nuggets” or solitons
1810.04360, 2105.02840

Macroscopic DM Searches

Even if R_{DM} is macroscopic, still hard to detect because of **rarity**



“Low” masses probed by terrestrial searches:

WIMP/neutrino detectors

1803.08044, 1812.09325

Etched plastic

2012.13406

Ancient mica

2105.06473

Gravitational wave detectors

1807.03788

Meteors and cosmic rays

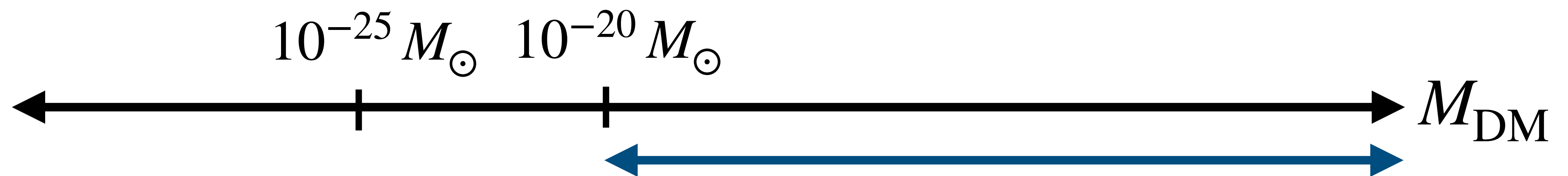
2008.01285

Seismic waves

1610.09680

Macroscopic DM Searches

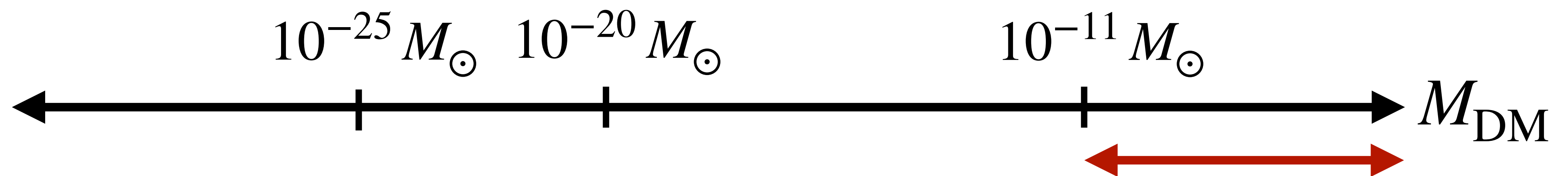
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For higher masses, too rare to ever detect by Earth impact, $\Gamma \lesssim 1/(10^4 \text{ yr})$

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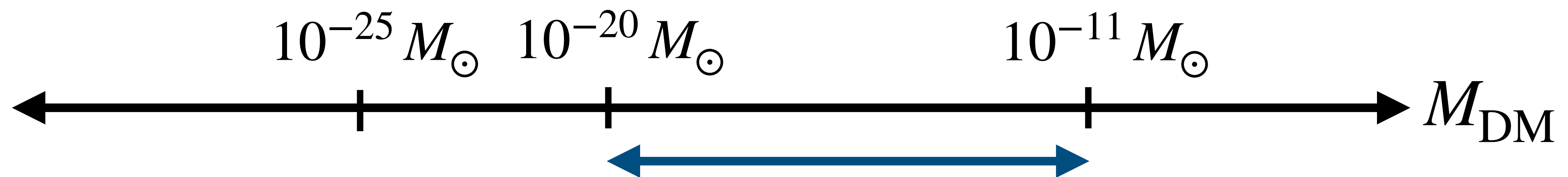


Highest masses excluded by gravitational lensing

1701.02151

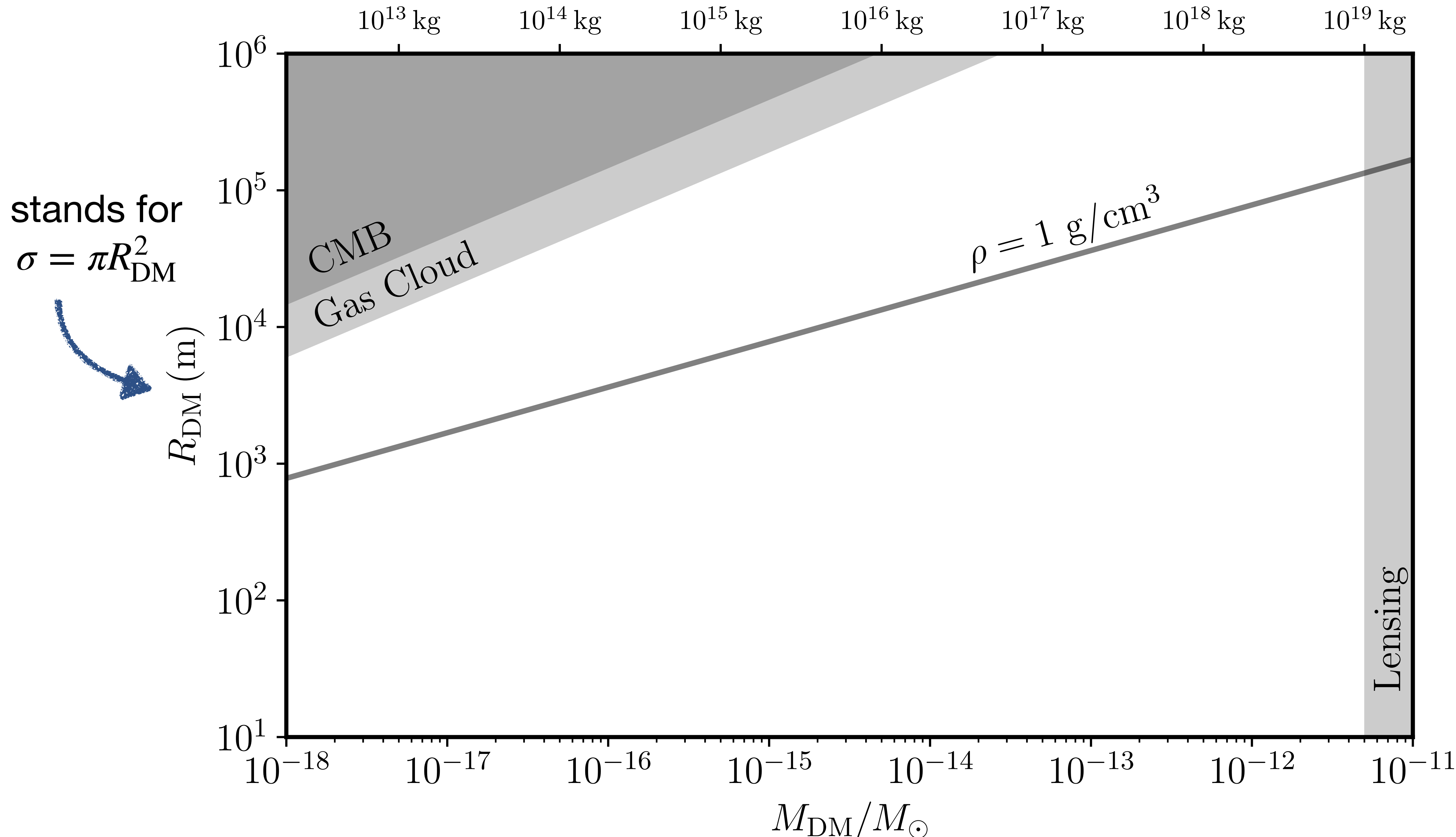
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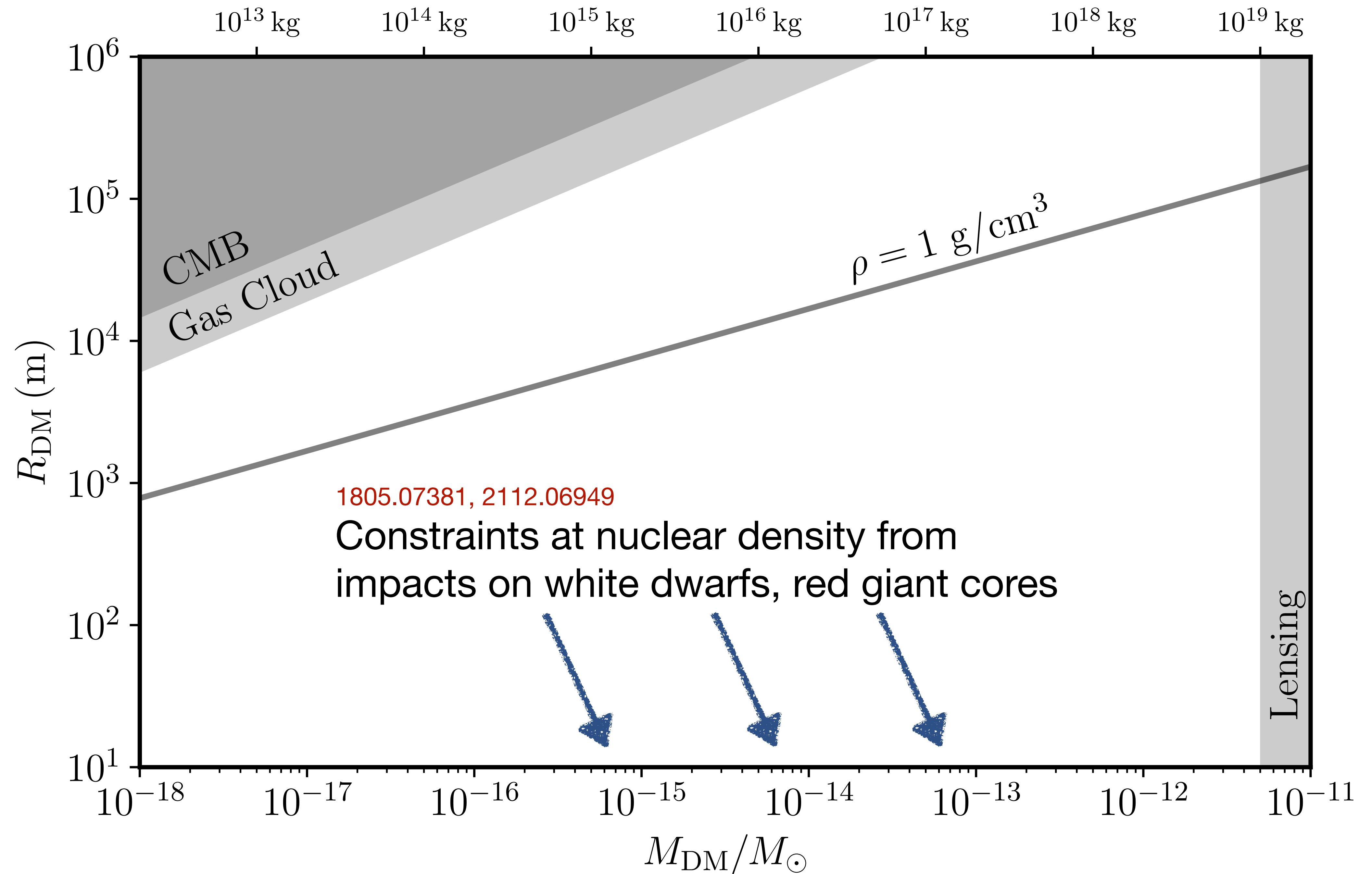
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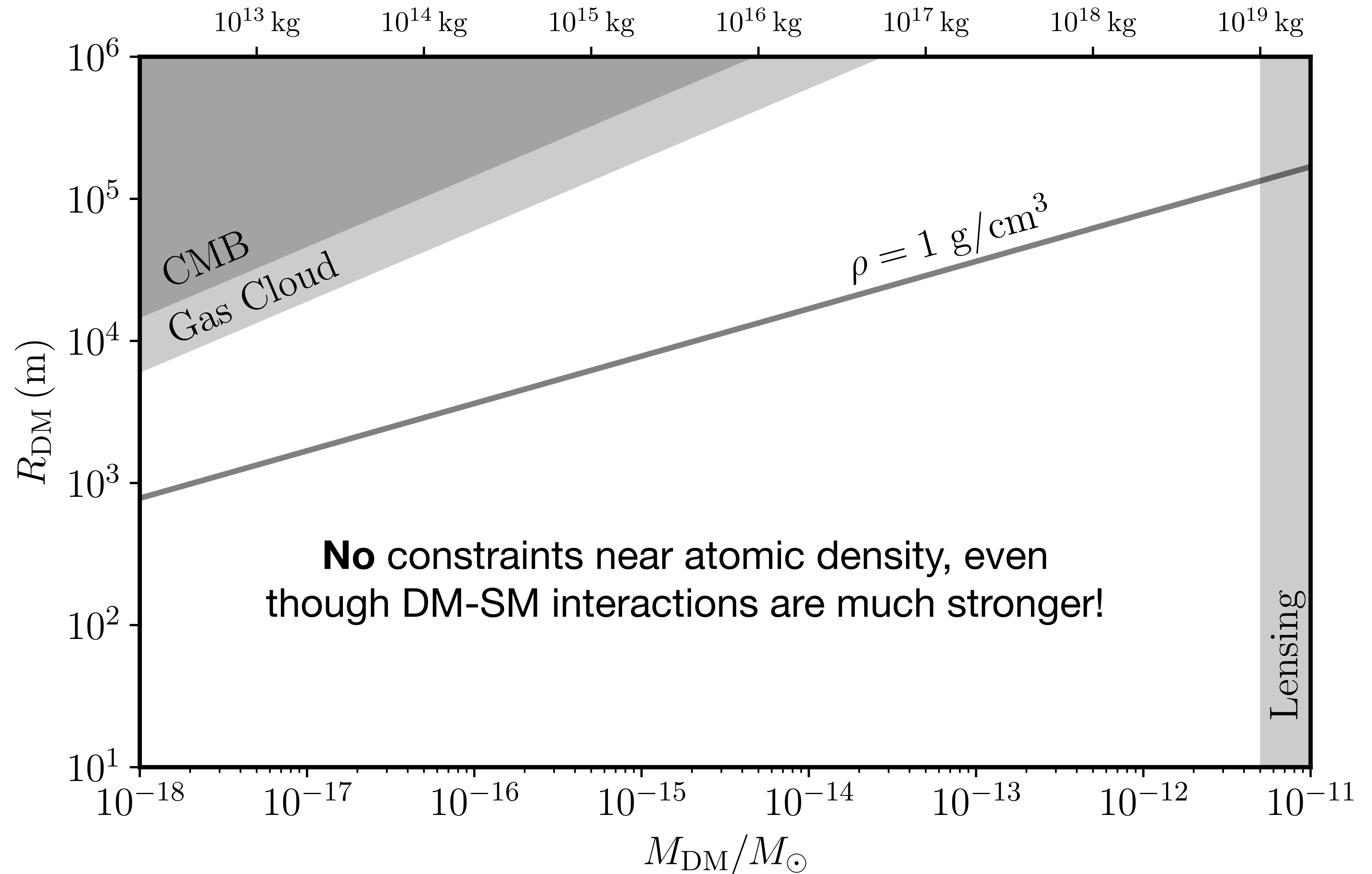


Leaves a relatively unconstrained “dark asteroid” range

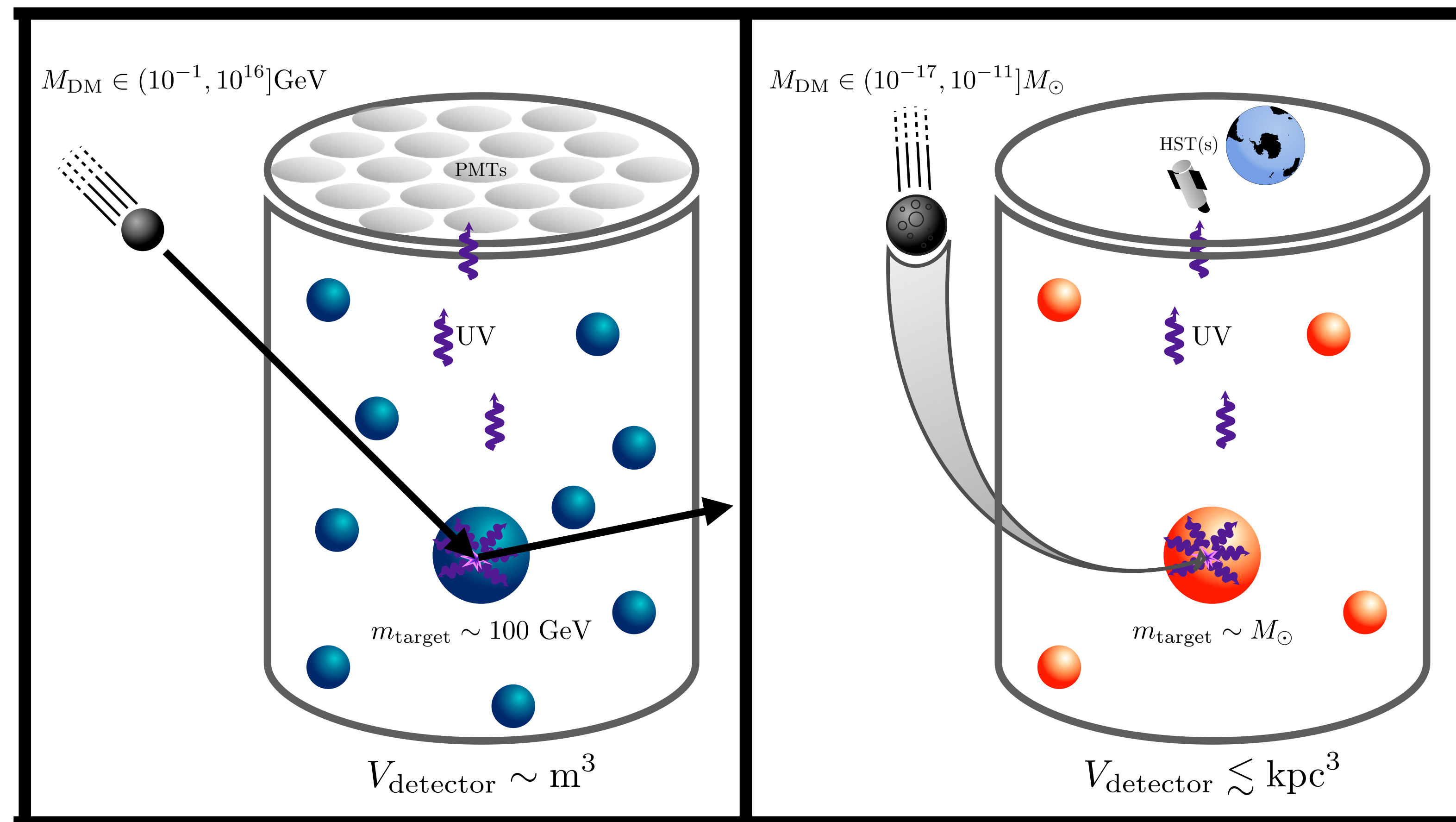
(Similar to asteroid window for primordial black holes, but differs in detail)







Probe by looking for **stellar impacts**: a dark matter direct detection experiment on astronomical scales!



Naive Estimates

Rate per star:

$$\Gamma \sim \frac{\rho_{\text{DM}} v_{\text{DM}}}{M_{\text{DM}}} \pi R_{\star}^2 \left(1 + \frac{v_{\text{esc}}^2}{v_{\text{DM}}^2} \right) \sim \frac{1}{10^4 \text{ yr}} \frac{10^{-15} M_{\odot}}{M_{\text{DM}}}$$

Energy dissipated in star:

$$E_{\text{DM}} \sim \frac{1}{2} M_{\text{DM}} v_{\text{esc}}^2 \sim 10^{34} \text{ erg} \frac{M_{\text{DM}}}{10^{-15} M_{\odot}}$$

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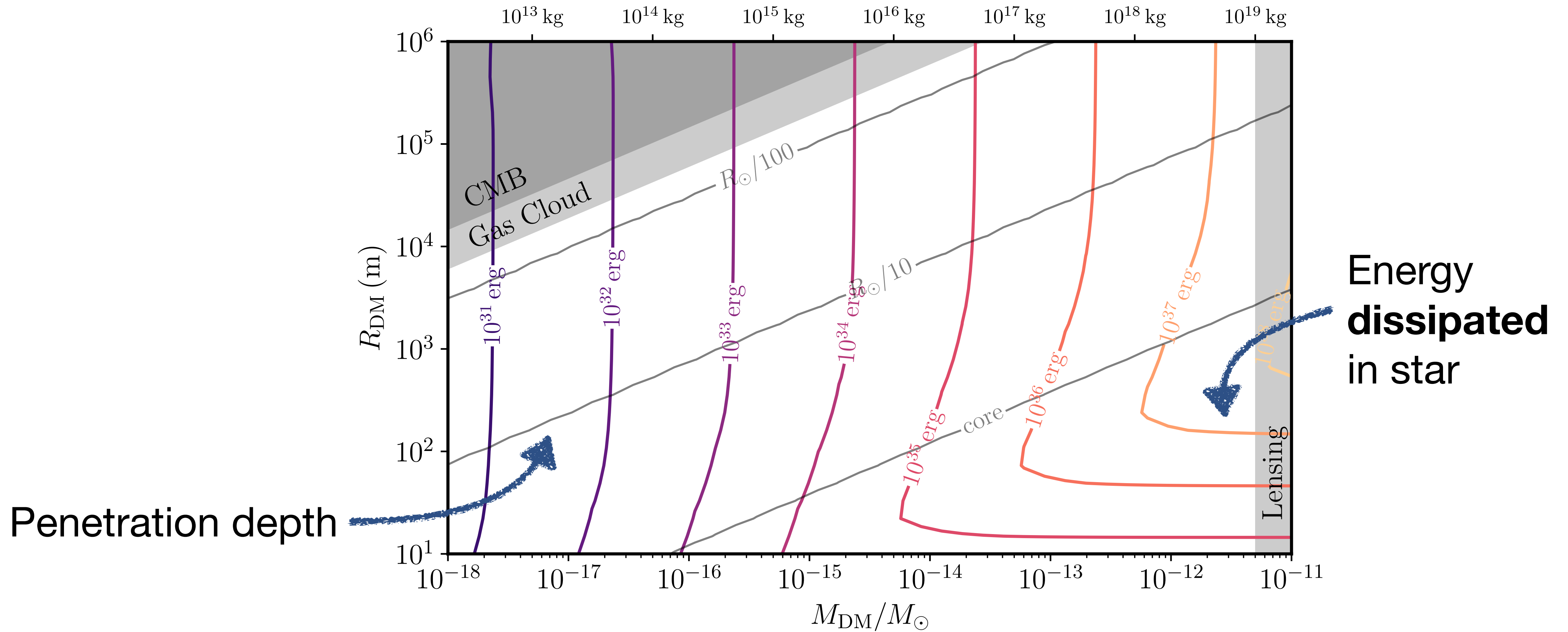
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Average power ΓE_{DM}
negligible, but E_{DM} high,
potentially observable as
strong transient



Dark asteroid experiences gravitational force $M_{\text{DM}}g$ and drag $F_d \sim \rho\sigma v^2$

Taking geometric cross section $\sigma = \pi R_{\text{DM}}^2$, but qualitatively similar for lower σ

Rigidity and Opacity

Ordinary solids are very weakly bound because of several small numbers:

$$\frac{E_b}{mc^2} \sim \frac{\text{intermolecular bond energy}}{\text{nuclear mass}} \sim \frac{\alpha^2 m_e / 10}{Nm_p} \sim 10^{-10}$$

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Asteroids fall apart in stellar atmosphere because energy dissipated is much larger:

$$\frac{E_k}{mc^2} \sim \frac{v_{\text{esc}}^2}{c^2} \sim 10^{-6}$$

Not hard to have weakly interacting dark sectors with $E_b/mc^2 \gtrsim 10^{-6}$, where dark asteroid remains intact throughout whole impact

Rigidity and Opacity

Geometric cross section does not need unreasonable DM-SM interactions, since dark asteroids are large:

$$\sigma \lesssim \frac{1}{n_{\text{DM}} R_{\text{DM}}} \sim \frac{m_{\text{DM}}}{\rho_{\text{DM}} R_{\text{DM}}} \sim (10^{-30} \text{ cm}^2) \frac{m_{\text{DM}}}{0.1 \text{ GeV}} \frac{1 \text{ g/cm}^3}{\rho_{\text{DM}}} \frac{1 \text{ km}}{R_{\text{DM}}}$$

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For light constituents, σ below direct detection bounds even if most DM is unbound

If DM-DM scattering also geometric, bounds on σ/m_{DM} weaker by orders of magnitude

How the Energy Escapes

Naively, energy deposited even at depth $R_{\star}/10$ takes **very** long time to get out

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But dark asteroid is traveling supersonically, at $\text{Ma} = v_{\text{esc}}/c_s \sim 100$, so energy dissipated forms shock waves, which efficiently propagate to surface!

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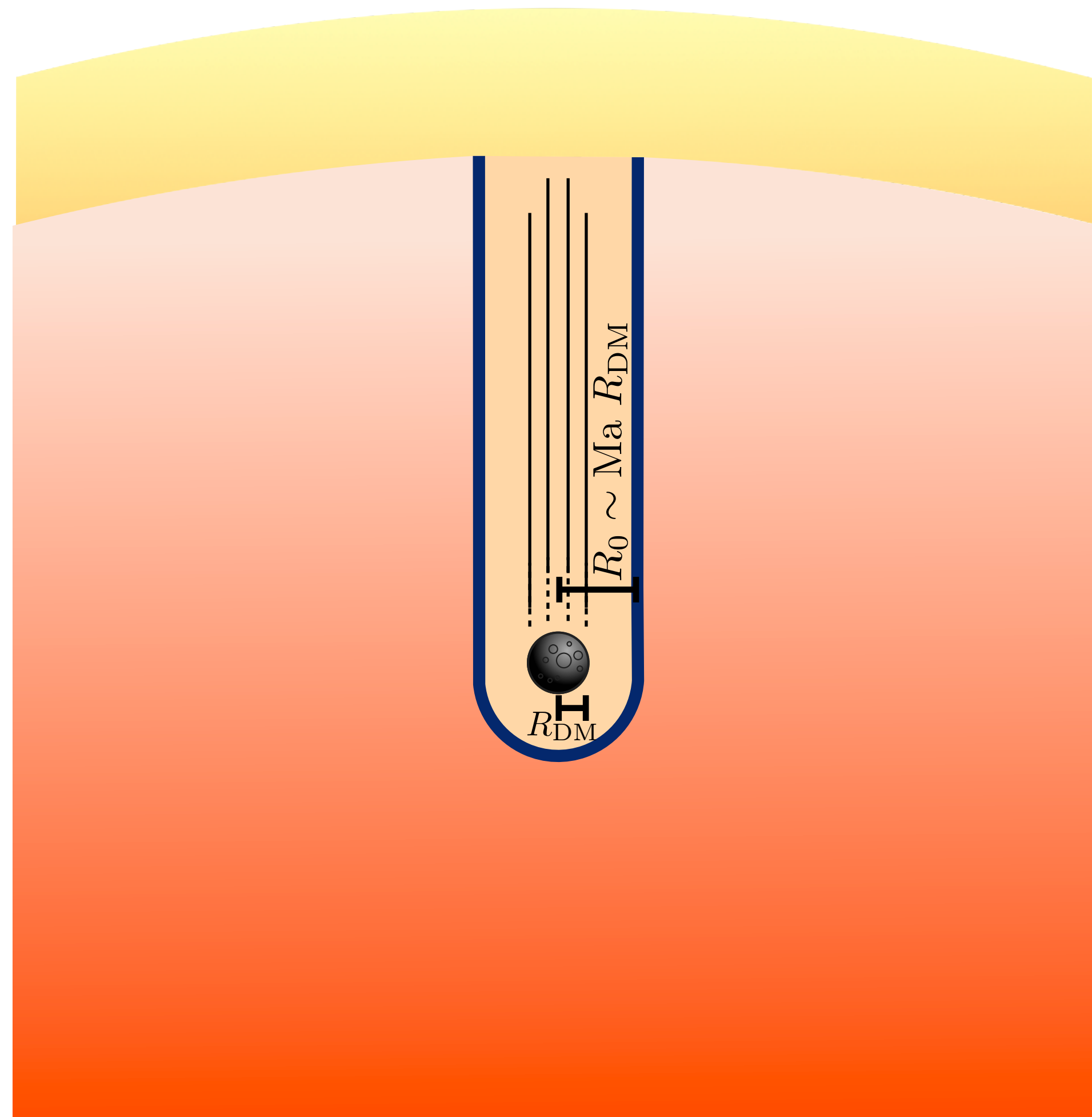
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A complicated hydrodynamic problem, but solvable with controlled approximations known for decades

Whitham (1956), ReVelle (1976)

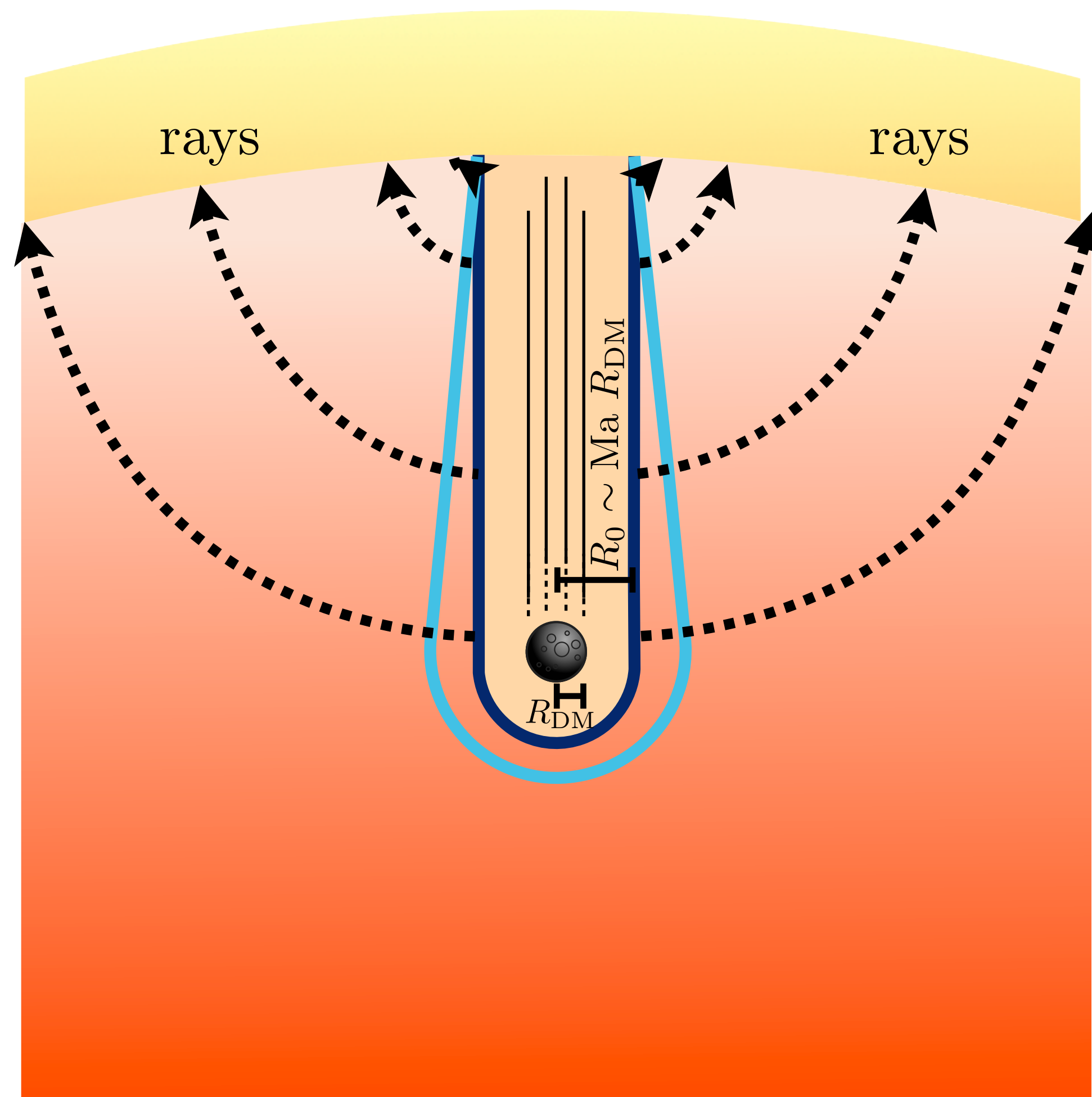
How the Energy Escapes



Near the dark asteroid: since $\text{Ma} \gg 1$, can approximate with known cylindrical blast wave

Characteristic size $R_0 \sim \text{Ma } R_{DM}$

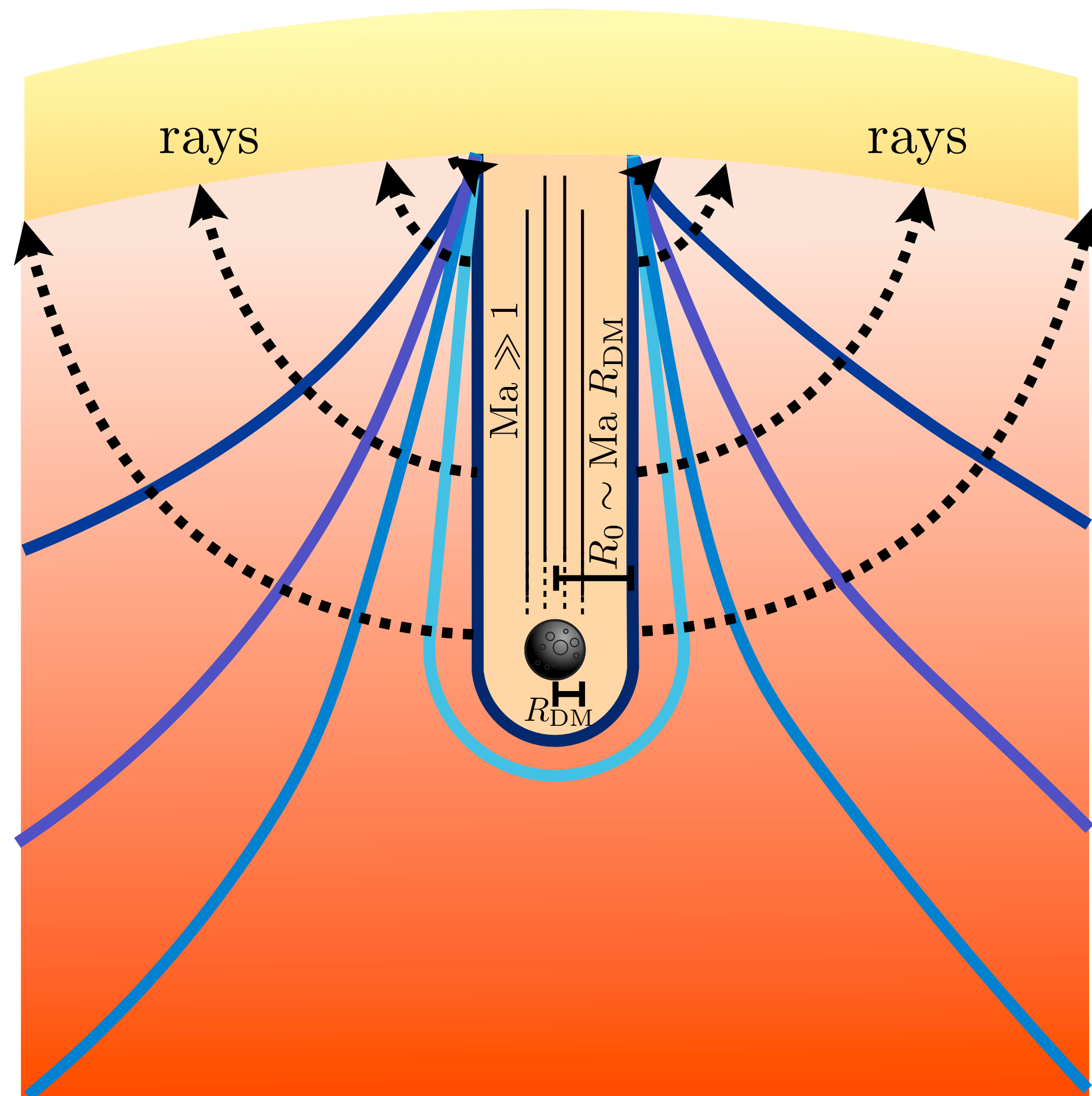
How the Energy Escapes



Beyond R_0 , match onto a weak shock solution — essentially an acoustic wave with additional dissipation

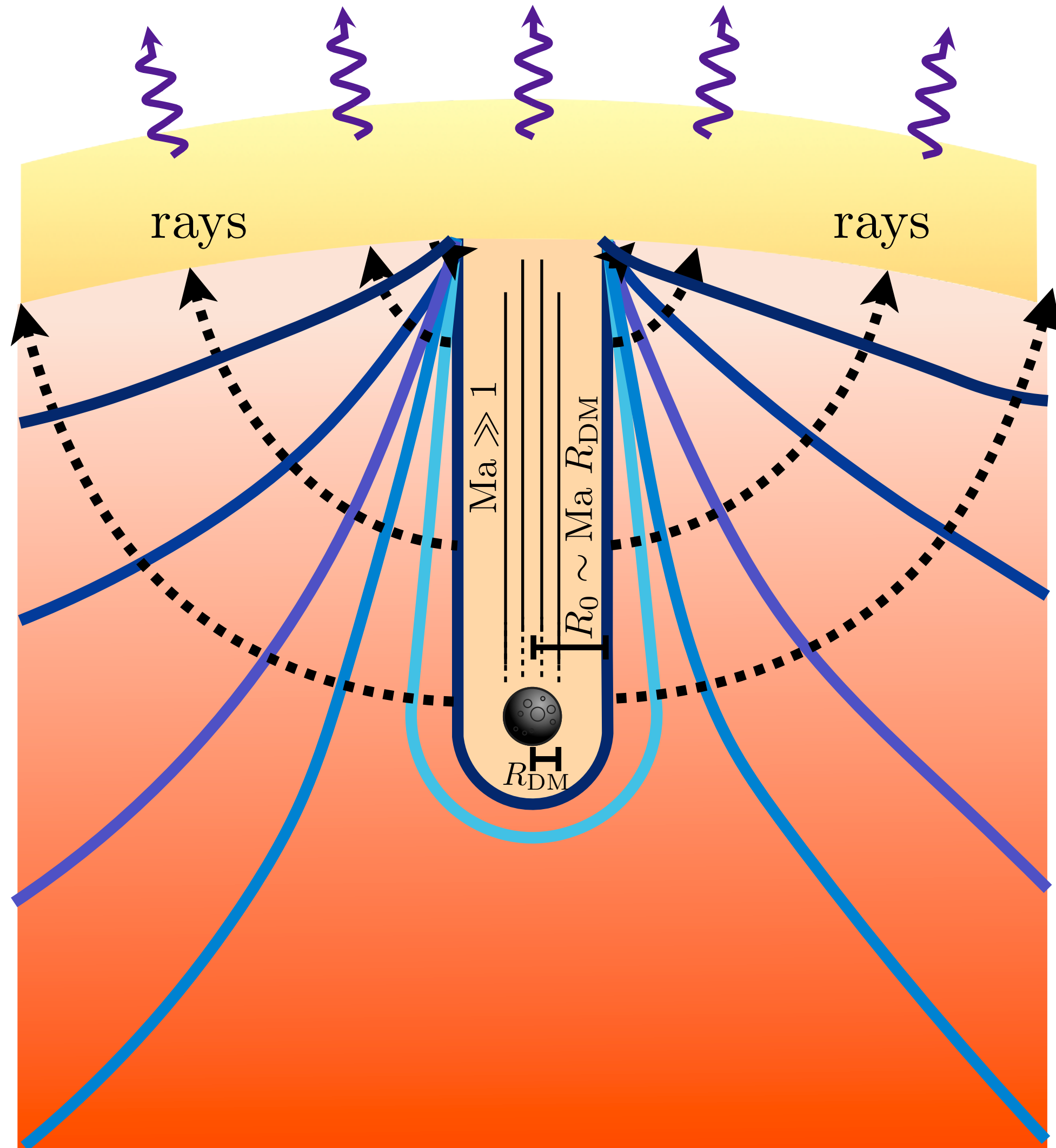
Rays refract radially outward

How the Energy Escapes

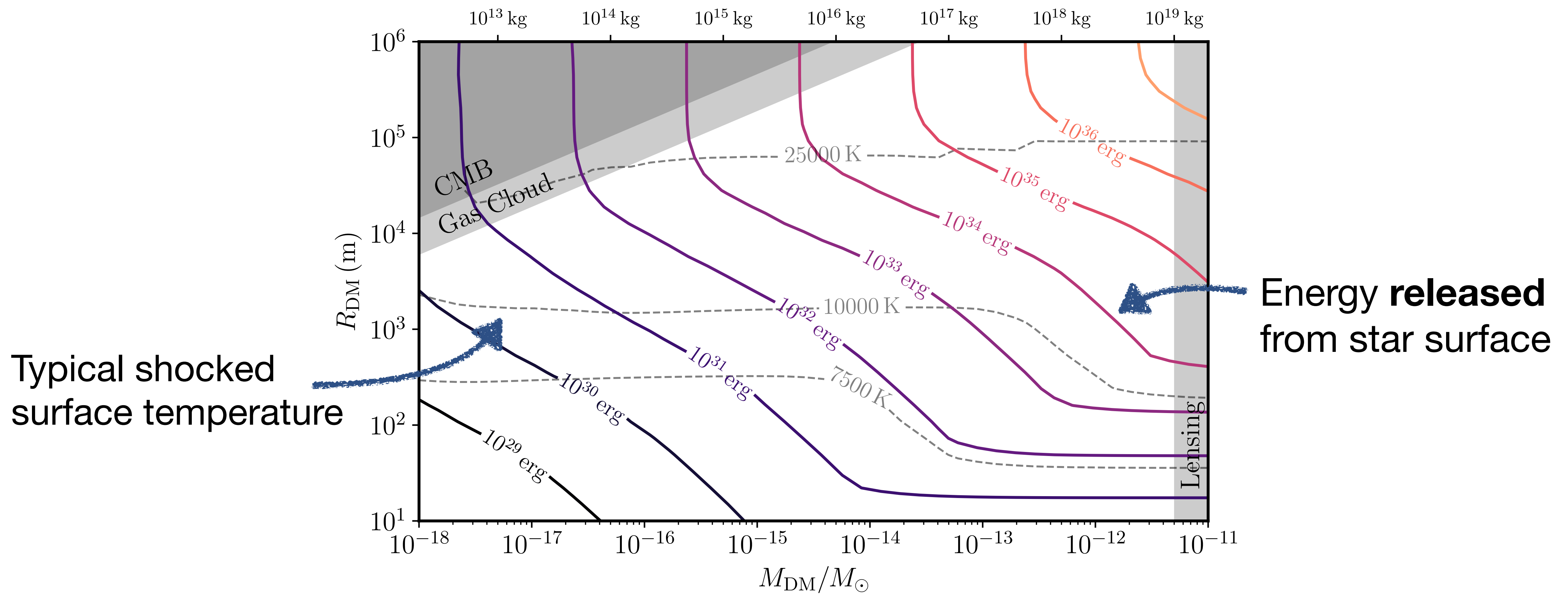


Weak shock waves lose energy, but grow in strength as they approach the stellar surface

How the Energy Escapes

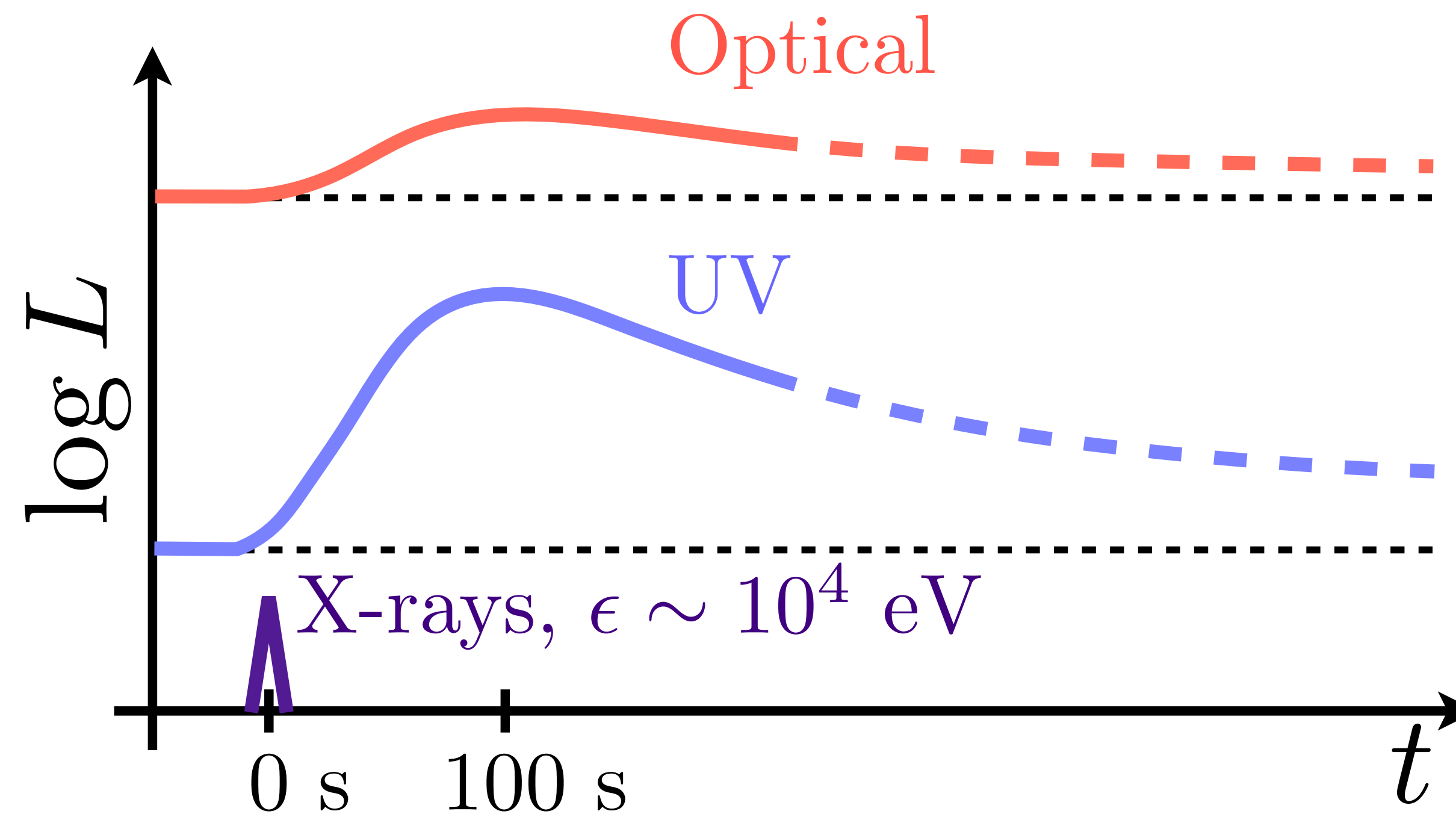


Shock waves grow strong very close to the photosphere, heating a patch of the surface



Point of shock calculation: typical final temperature is UV; energy released is suppressed by shock dissipation as R_{DM} decreases, but still sizable

Detecting the Signal



Main signature: thermal UV transient spread over ~hundreds of seconds

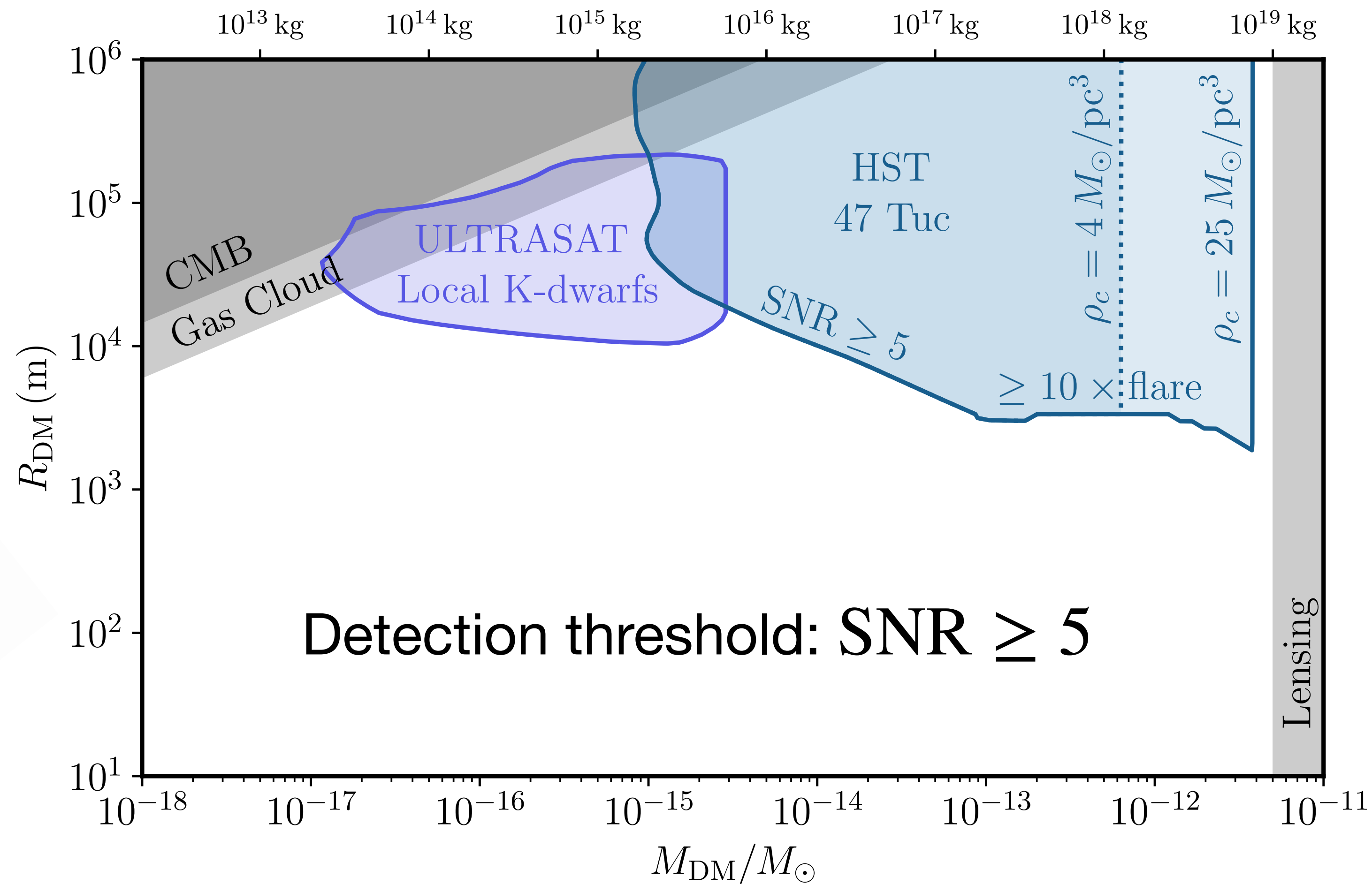
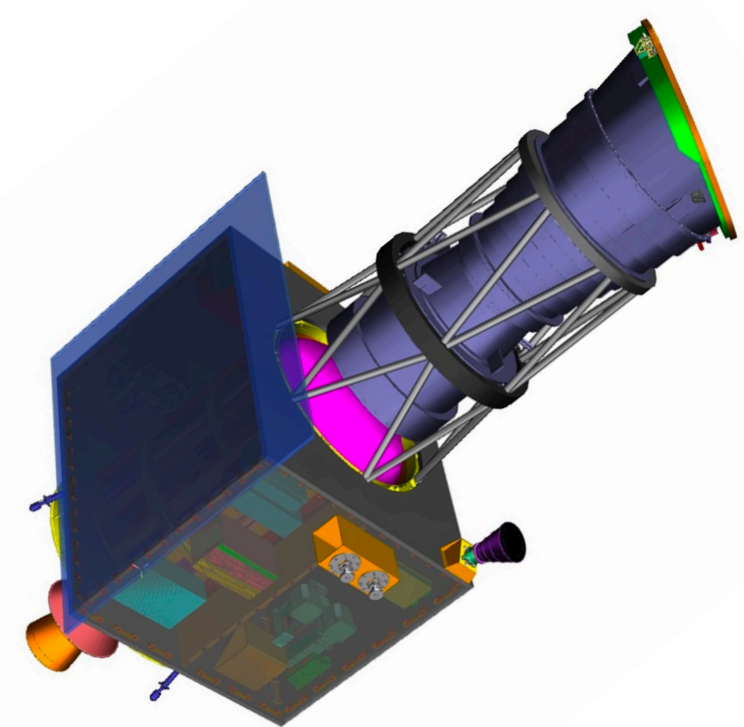
Search Strategies

Local search

- Wide field search for transients on nearby (within 1 kpc) stars
- Does not require dedicated survey
- Rate low, so targets low M_{DM}
- Stellar superflare background could be high

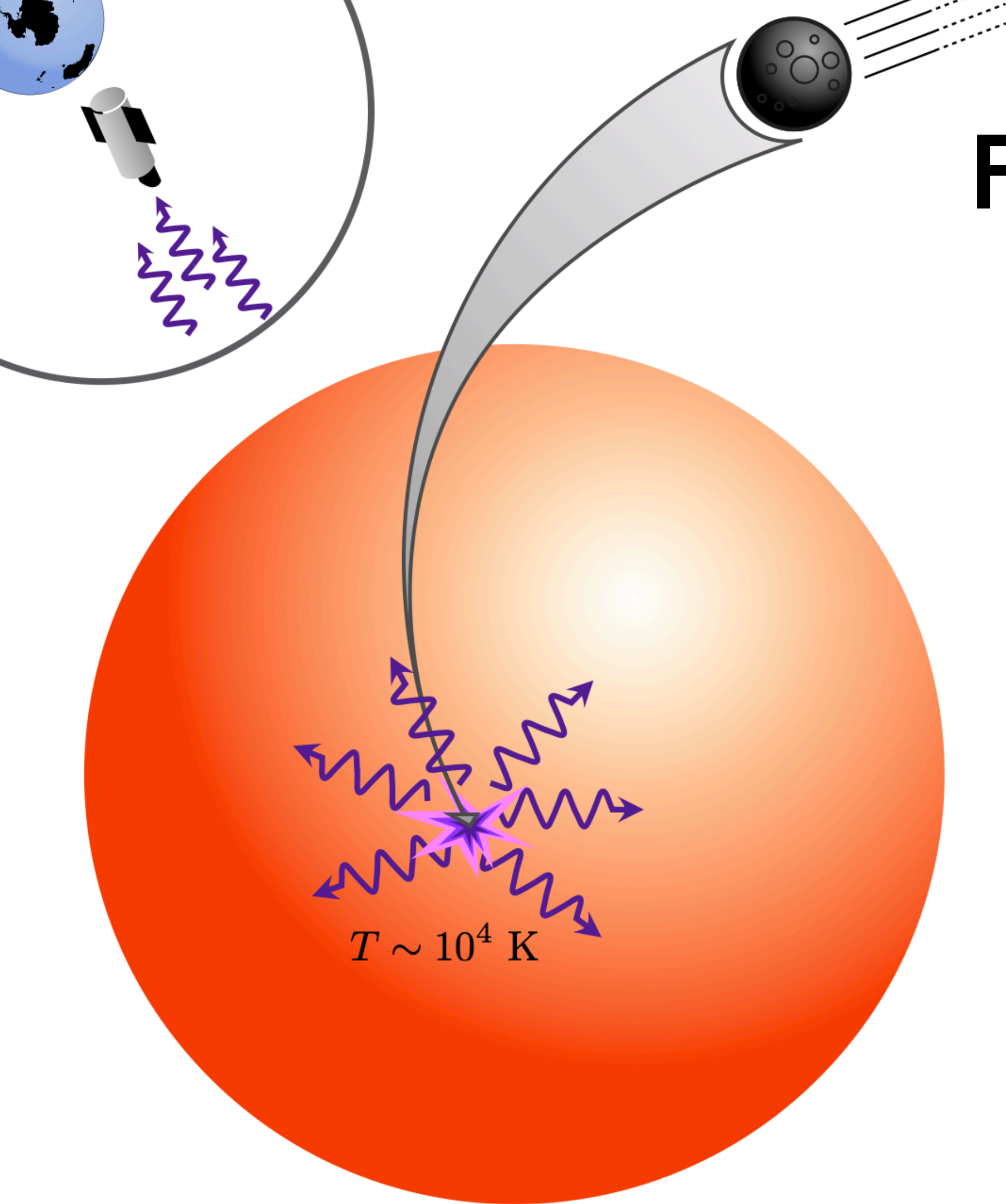
Focused search

- Point to region with high $\Gamma \propto \rho_{\text{DM}}/v_{\text{DM}}$
- Requires dedicated search with powerful UV telescope
- Multiple kpc away, so targets high M_{DM}
- Events very energetic, background negligible



Local search: planned UV transient telescope ULTRASAT, nearby K dwarfs

Focused search: Hubble Space Telescope, core of globular cluster 47 Tuc



Future Directions

- Constraints with archival or future HST data
- Other star types (brown dwarf), telescopes (LSST), or locations (galactic center)
- Full simulation of shock propagation and radiative cooling

Plenty of low-hanging fruit!

Popular Interest

Dark matter travelling through stars could produce potentially detectable shock waves

"Dark matter asteroids" could be spotted colliding with stars

Des « astéroïdes » de matière noire se révéleraient lors de collisions avec des étoiles

La materia oscura podría generar ondas capaces de ser detectadas

Раскрыты последствия столкновения темной материи со звездой

ASTEROIDEN AUS DUNKLER MATERIE (SO FERN VORHANDEN) KÖNNEN SONNENERUPTIONEN VERURSACHEN

GOT YOU!
SCIENTISTS MAY HAVE FOUND A STRANGE WAY TO DETECT DARK MATTER COLLISIONS

Subject of a brief popsci news cycle (strange experience)

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SCIENTISTS MAY HAVE FOUND A STRANGE WAY TO DETECT DARK MATTER COLLISIONS

Would be amazing if dark matter had always been easily observable if you looked for the right thing. Also: bad news and major potential hazards for future starships.

If I look up, will I see the dark matter asteroid coming to destroy earth ?

Only in theaters and even then only when Roland Emmerich gets around to it.

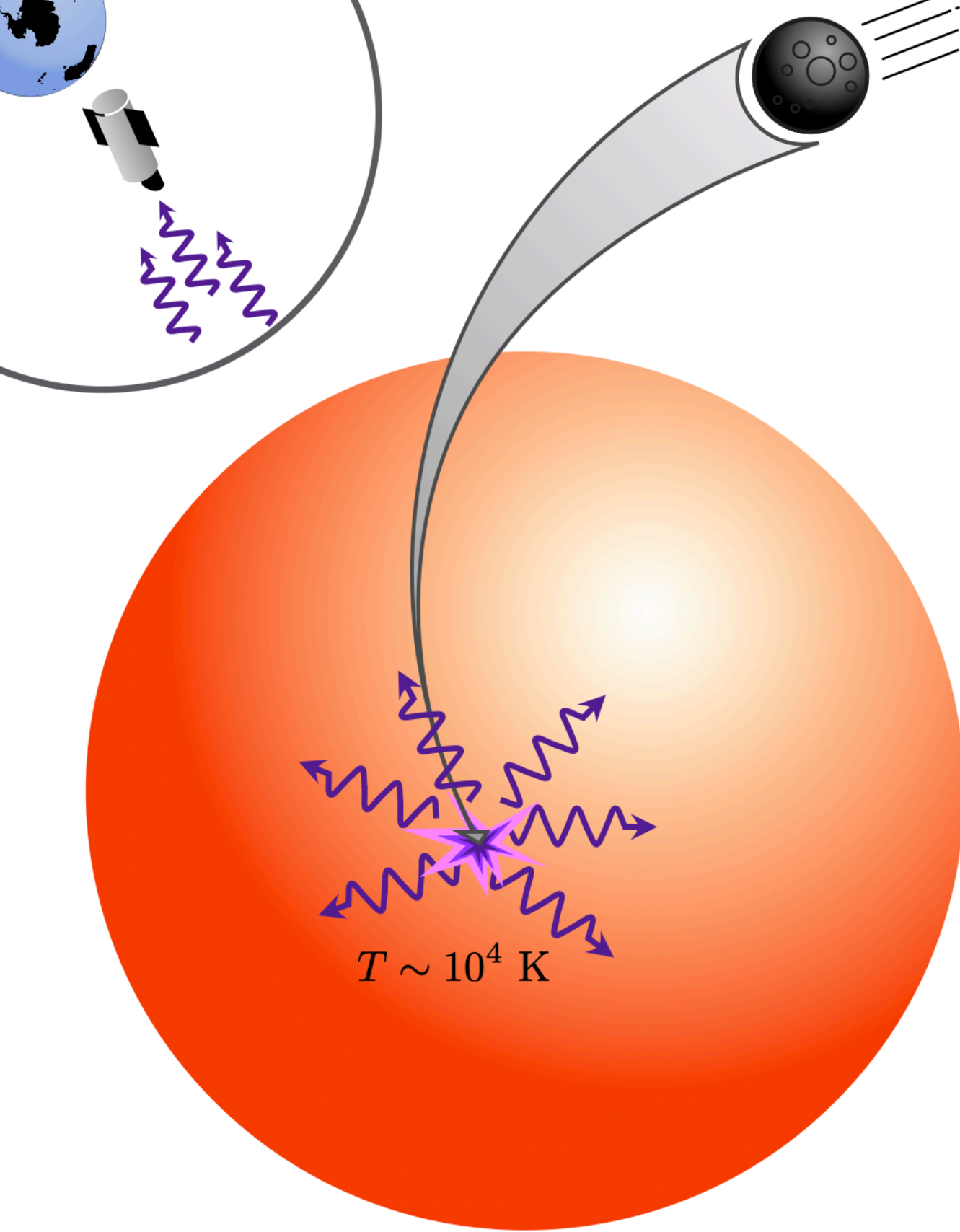
I entirely agree with both you and the article.

This stuff is hard to find, so we need to look for it in every way we can think of.

Something's there, and we need to find it.

Many people care about our field, and clear communication is important!

Outline

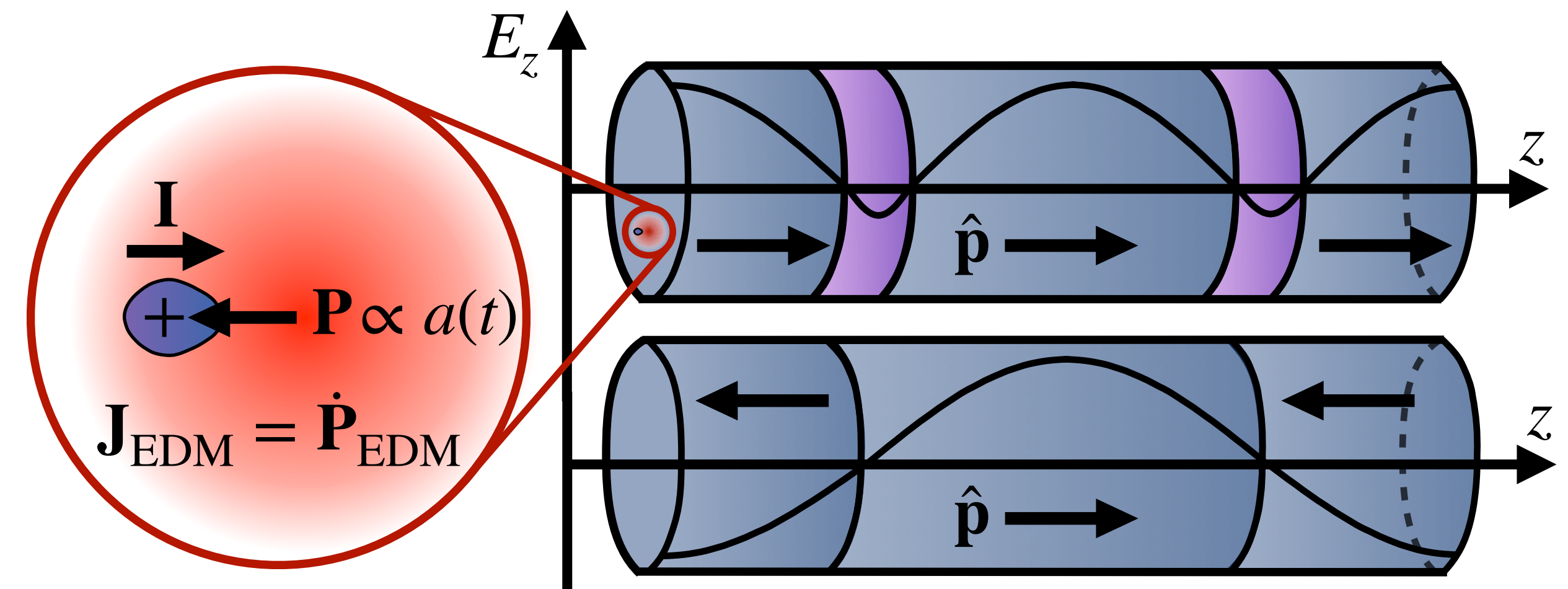


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The first search for the QCD axion's defining coupling at GHz frequencies

Berlin, KZ (in prep)



The QCD Axion

A pseudoscalar field a with defining coupling to gluons

$$\mathcal{L} \supset \theta_a \frac{\alpha_s}{8\pi} G^{\mu\nu} \tilde{G}_{\mu\nu} \quad \theta_a = \frac{a}{f_a}$$

QCD effects yield potential minimum at CP-conserving point, and mass

$$m_a = 5.7 \mu\text{eV} \frac{10^{12} \text{ GeV}}{f_a}$$

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Key appeal: QCD axion dynamically relaxes CP violation over cosmological time

And for initial $\theta_a \sim 1$, residual oscillations are dark matter if $m_a \sim (0.5 - 50) \mu\text{eV}$

Detecting the QCD Axion

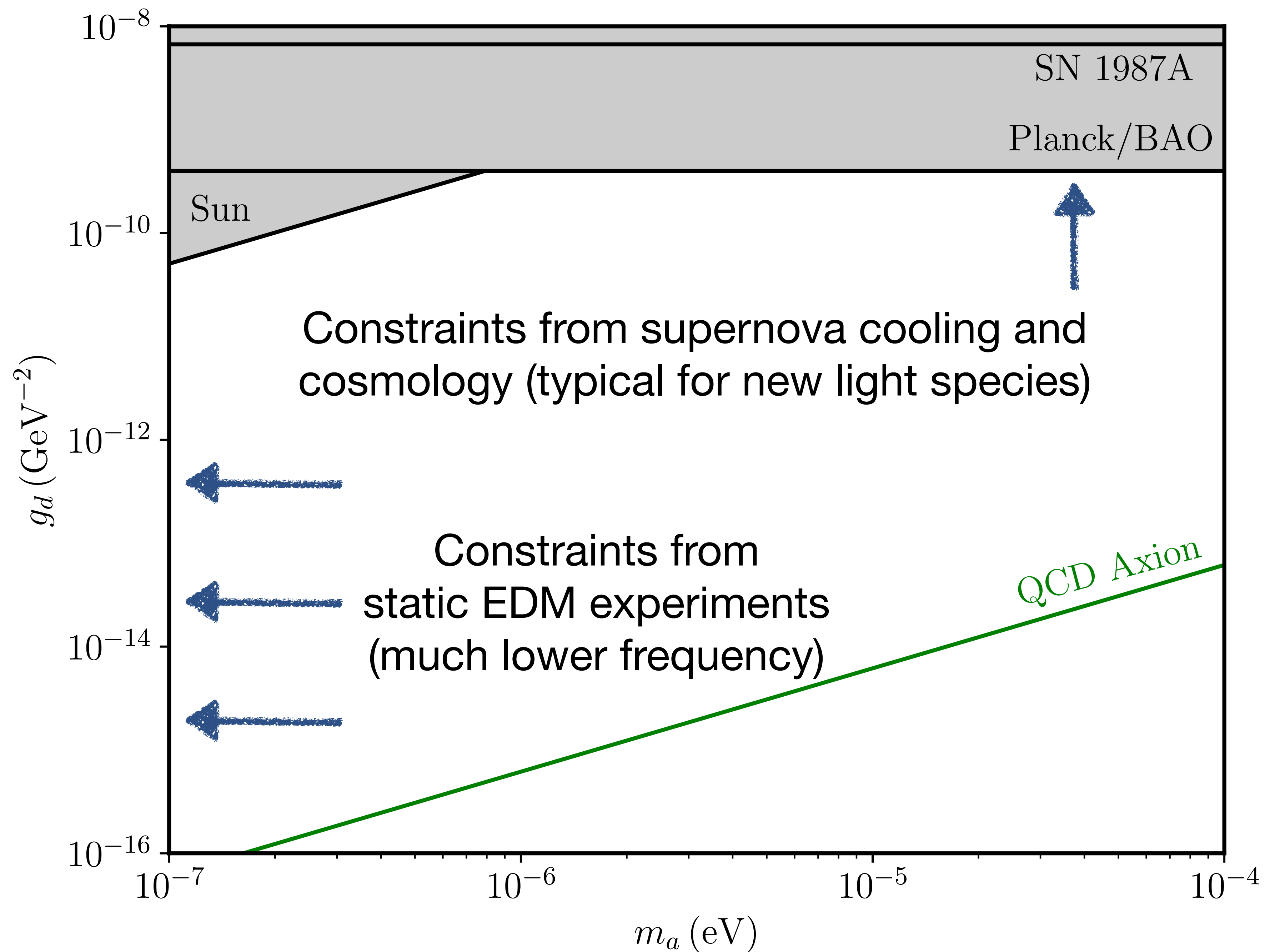
The QCD axion has small residual oscillations about the CP conserving point

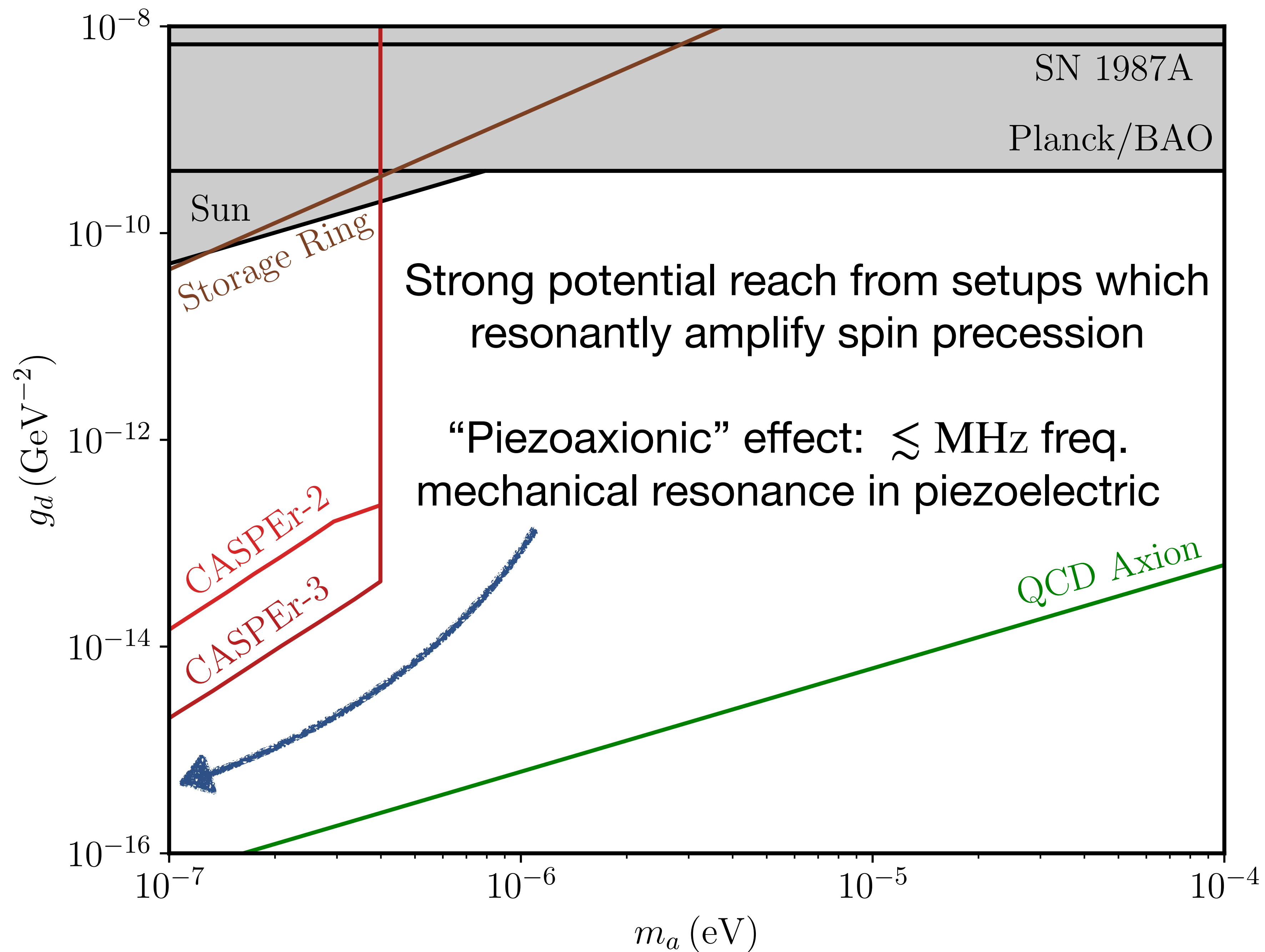
$$\theta_a(t) \simeq \frac{\sqrt{2\rho_{\text{DM}}}}{m_a f_a} \cos m_a t = 4 \times 10^{-19} \cos m_a t$$

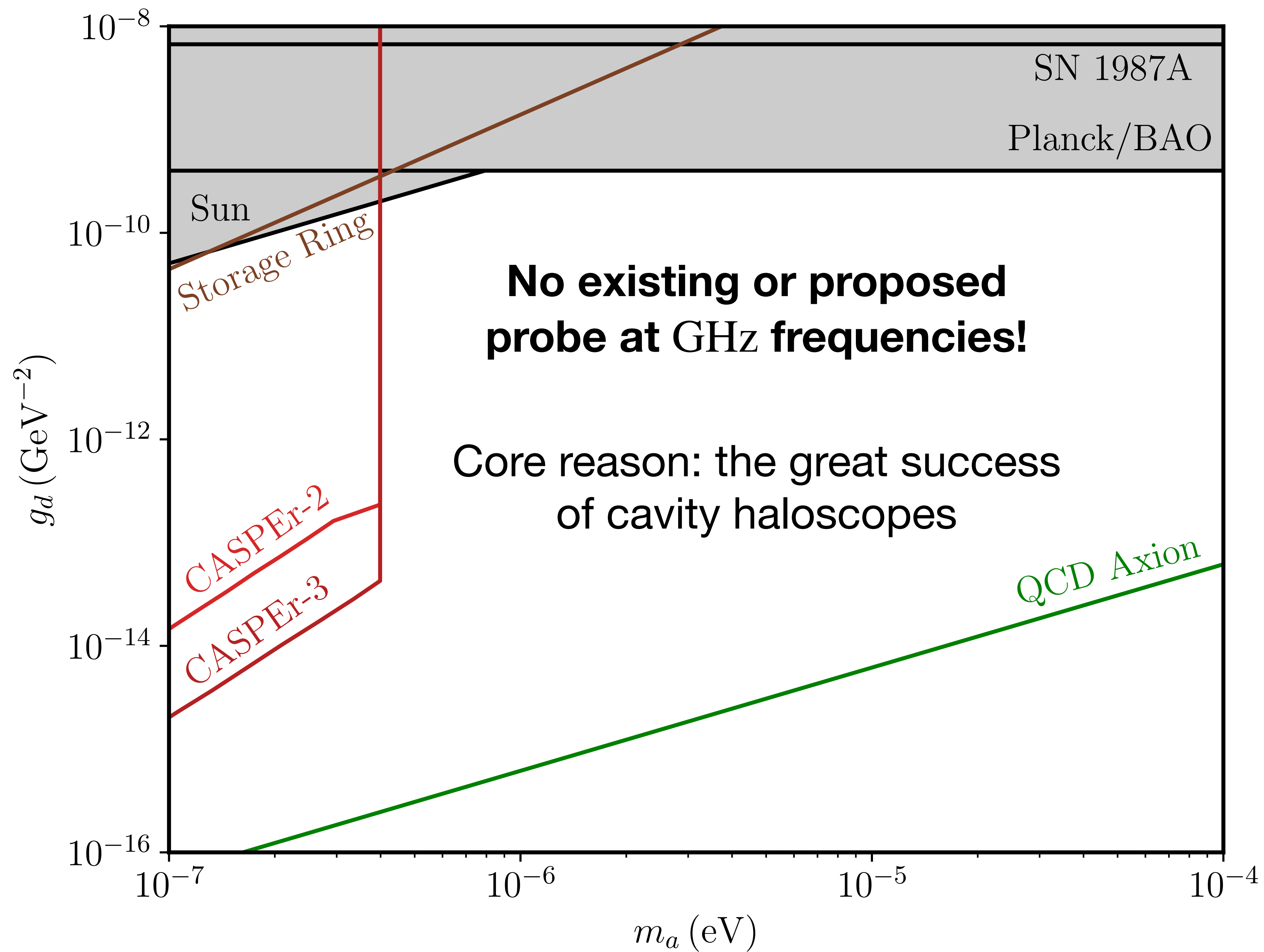
Its defining signature is oscillating CP violating nuclear effects, like neutron EDM

$$d_n = (2.4 \times 10^{-3} \text{ e fm}) \theta_a \equiv g_d a$$

Tiny effect hard to measure, especially at GHz axion frequencies ($m_a \sim \mu\text{eV}$)







Electromagnetic Axion Detection

Generically, QCD axions have a coupling to electromagnetism

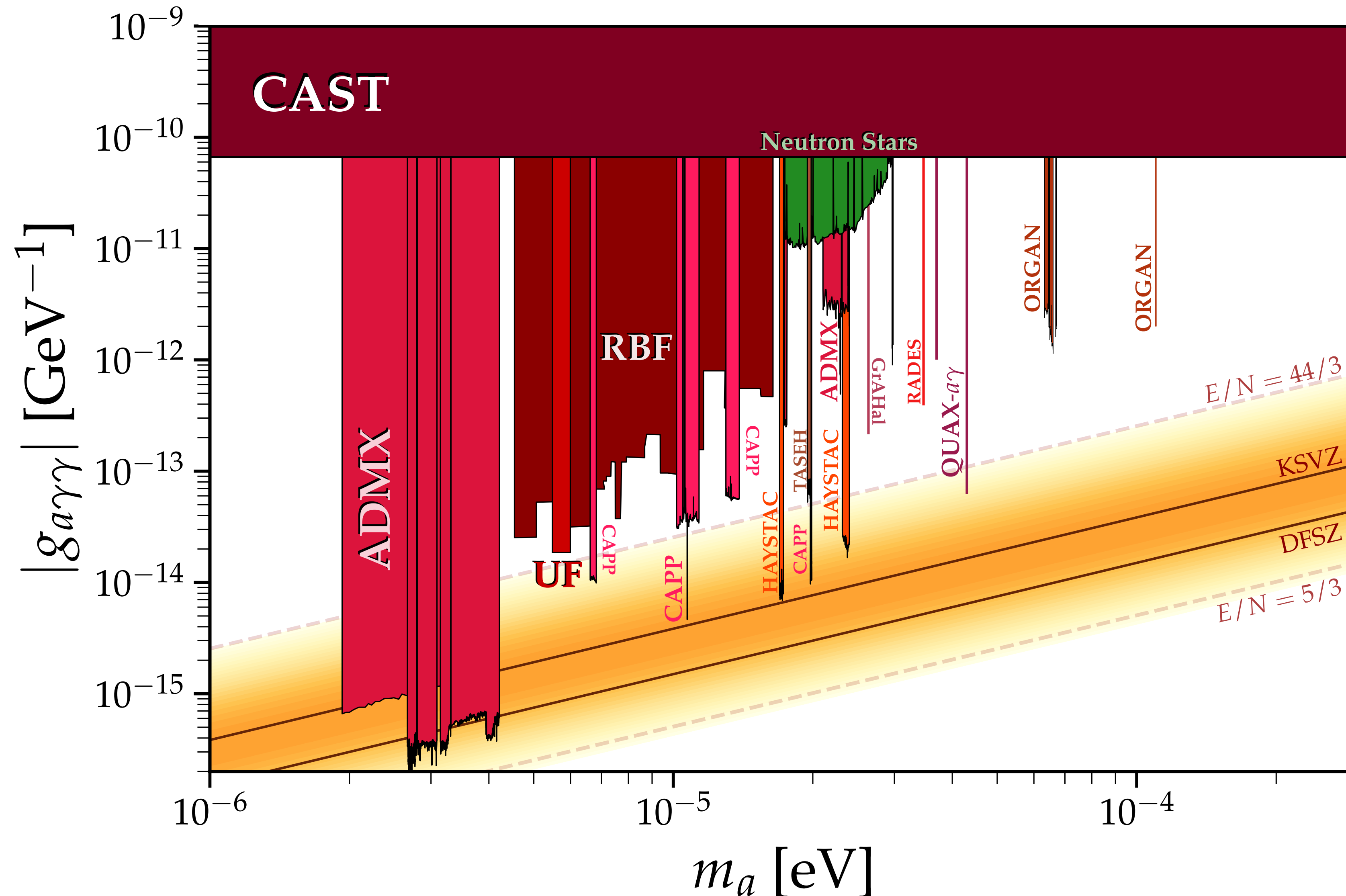
$$\mathcal{L} \supset -\frac{1}{4}g_{a\gamma\gamma} a F_{\mu\nu}\tilde{F}^{\mu\nu}, \quad g_{a\gamma\gamma} = \frac{\alpha C_\gamma}{2\pi f_a}$$

In the presence of a magnetic field, produces effective current $\mathbf{J}_{\text{eff}} = g_{a\gamma\gamma}\dot{a}\mathbf{B}$

Cavity haloscope: current resonantly excites microwave cavity mode with $\omega = m_a$

For typical laboratory-scale cavities, naturally targets GHz frequencies

The Cavity Haloscope



- Steady improvement, 35 years of development
- In past few years, sensitive to canonical QCD axion models

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- In past few years, sensitive to canonical QCD axion models
- Rapidly growing field, with wide international interest

Can translate this success to the axion-gluon coupling with **polarization haloscopes**

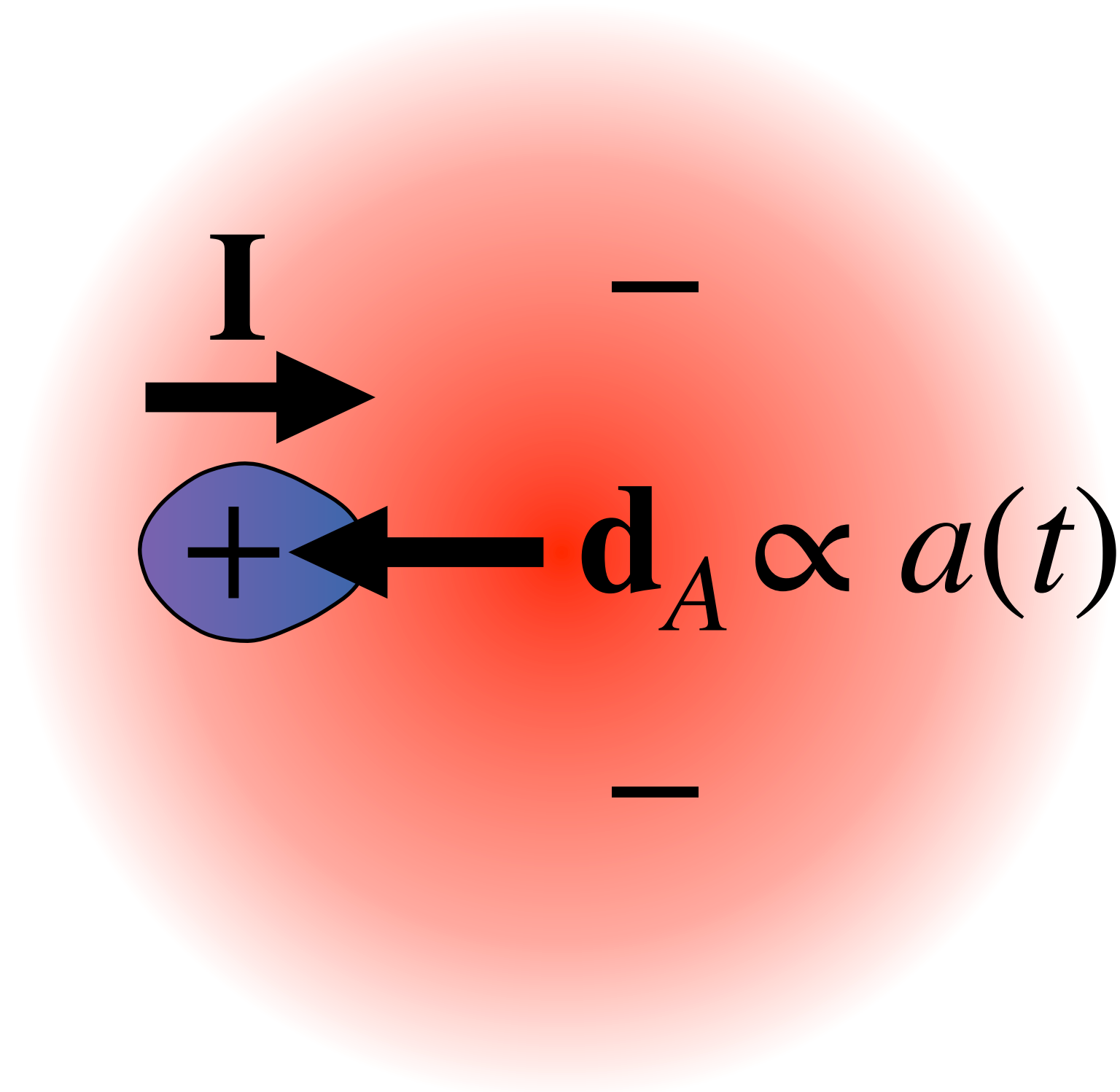
Polarization Currents

- The QCD axion produces atomic EDM $\mathbf{d}_A$ parallel to the nuclear spin $\mathbf{I}$
- For a nuclear spin-polarized sample, gives polarization

$$\mathbf{P}_{\text{EDM}} = n_A \mathbf{d}_A$$

- Time variation leads to current

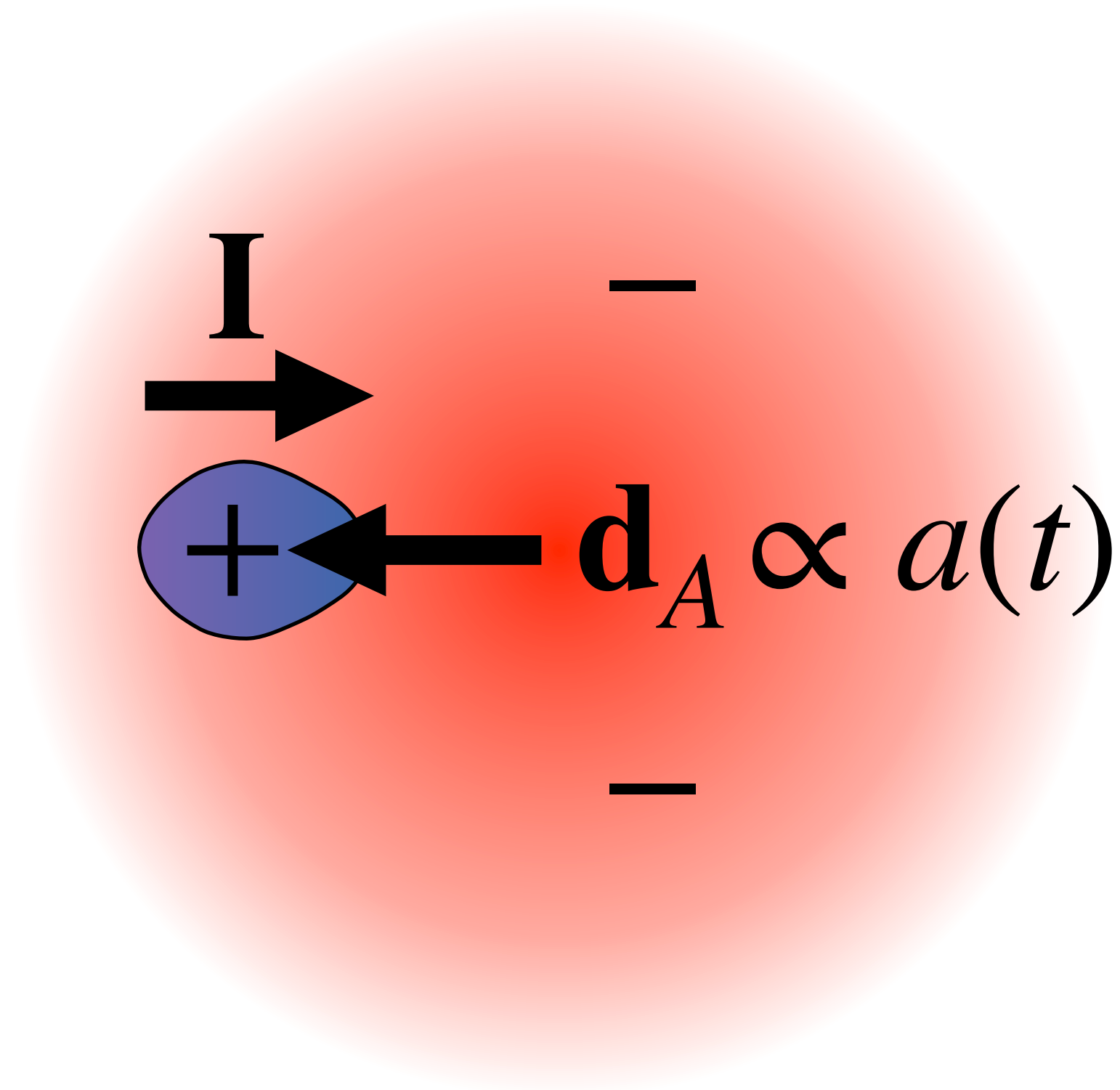
$$\mathbf{J}_{\text{EDM}} = \dot{\mathbf{P}}_{\text{EDM}}$$



Polarization Currents

To make a polarization haloscope, fill a cavity with nuclear spin-polarized dielectric, yielding

$$\mathbf{J}_{\text{EDM}} = g_d \dot{a} n_A \frac{d_A}{d_n}$$



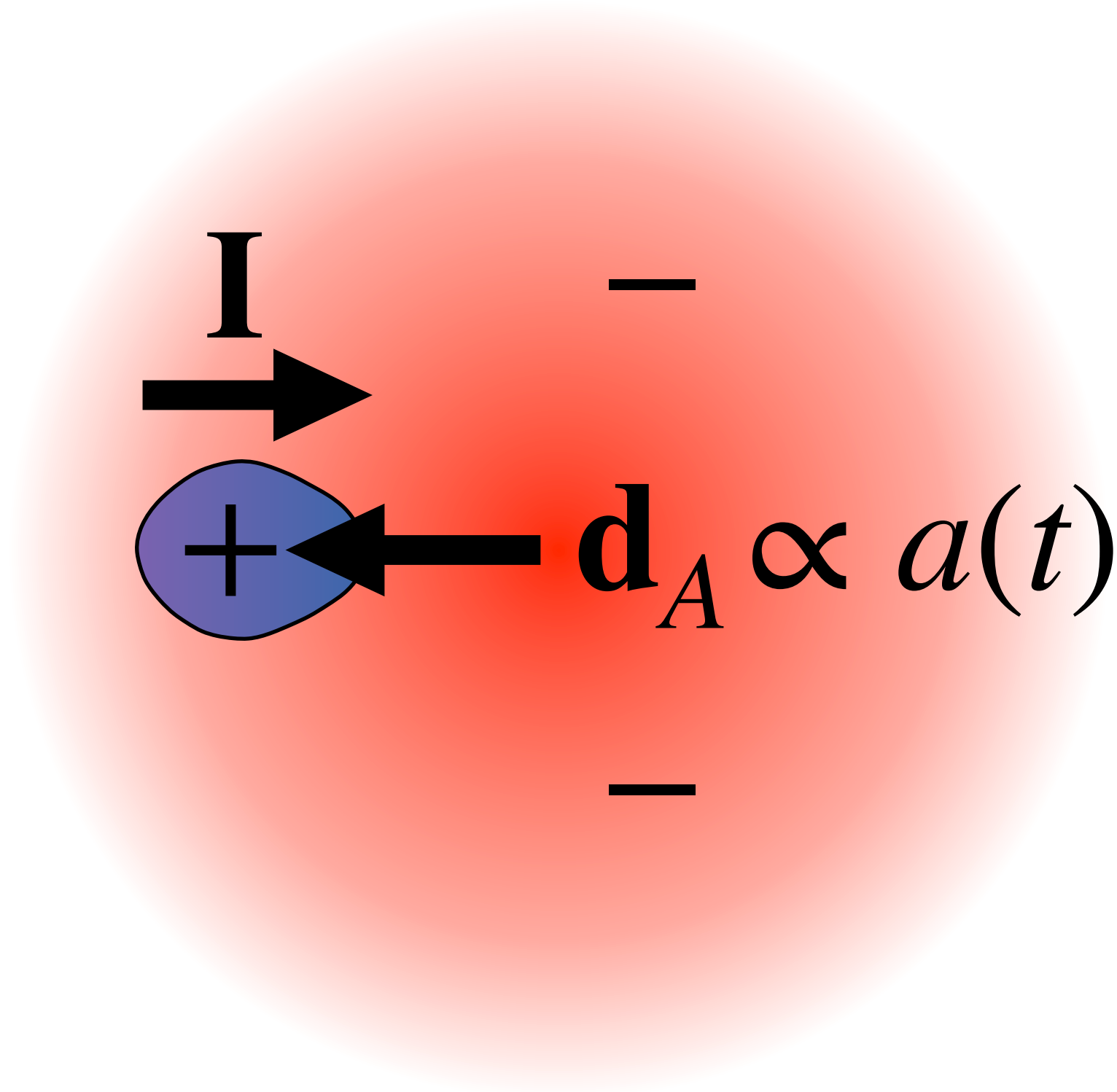
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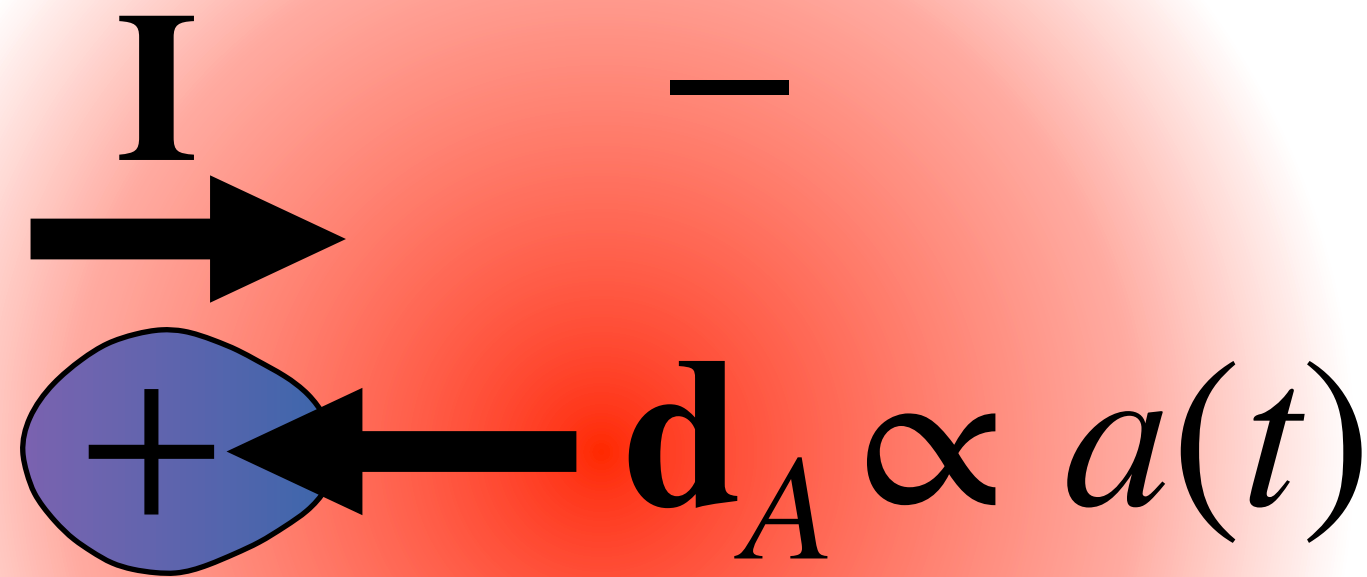
For a typical QCD axion:

$$\frac{J_{\text{EDM}}}{J_{\text{eff}}} \simeq 10^{-3} \frac{d_A}{d_n} \frac{n_A}{5 \times 10^{22} \text{ cm}^{-3}} \frac{8 \text{ T}}{B}$$



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Harder to reach QCD axion this way; why do it?

- Probes non-QCD axions with CP violating couplings
- Removes model-dependence (photon coupling can easily be 10 times lower)
- Only known way to verify a cavity haloscope discovery is the QCD axion

Signal Power

Analogous to cavity haloscope: for mode with profile $\mathbf{E}_i$ on resonance ($m_a \simeq \omega_i$),

$$P_{\text{sig}} \simeq m_a (f_p n_0 d_A)^2 \frac{V}{\bar{\epsilon}} \eta_i^2 \min(Q_a, Q_i)$$

Density of polarized
spins (fraction f_p)

Amplitude of
induced atomic EDM

Resonant
enhancement

($d_A \sim d_n$ for octupole deformed
nuclei, e.g. ^{161}Dy , ^{153}Eu)

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Density of polarized
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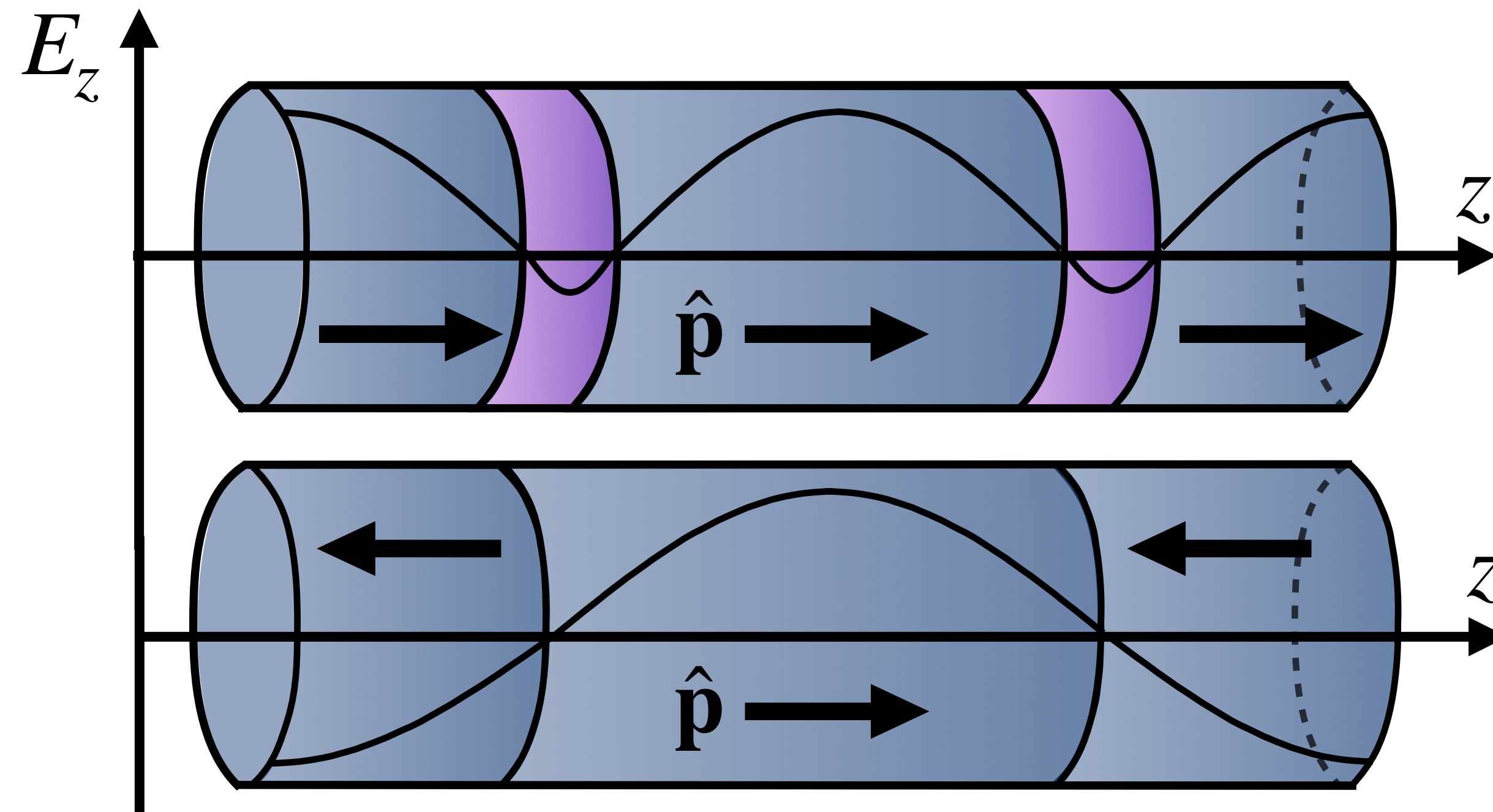
Geometric form factor

($d_A \sim d_n$ for octupole deformed
nuclei, e.g. ^{161}Dy , ^{153}Eu)

$$\eta_i^2 = \frac{\bar{\epsilon}}{V} \frac{|\int_{V_p} d^3\mathbf{x} \mathbf{E}_i \cdot \hat{\mathbf{p}}|^2}{\int_V d^3\mathbf{x} \epsilon E_i^2} \lesssim 1$$

Optimizing Geometry

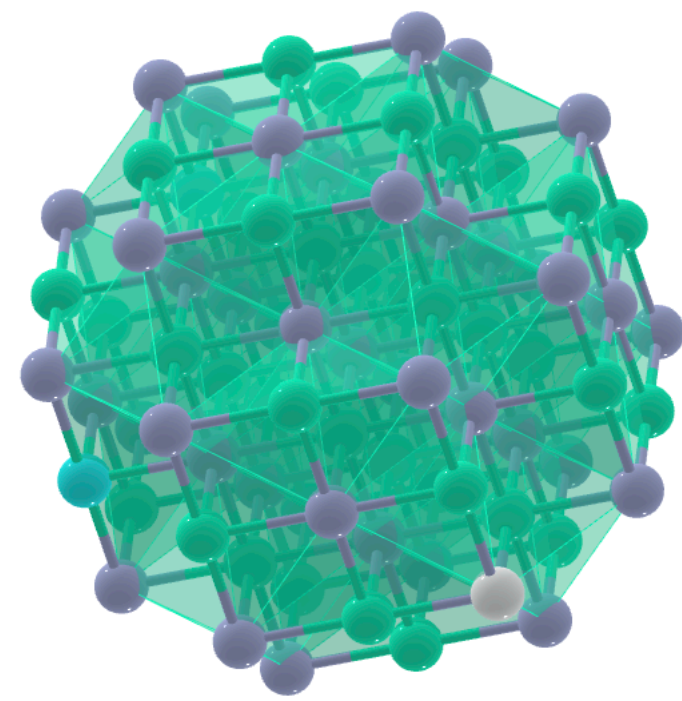
To maximize the geometric form factor, align nuclear spins $\hat{\mathbf{p}}$ with $\mathbf{E}_i$



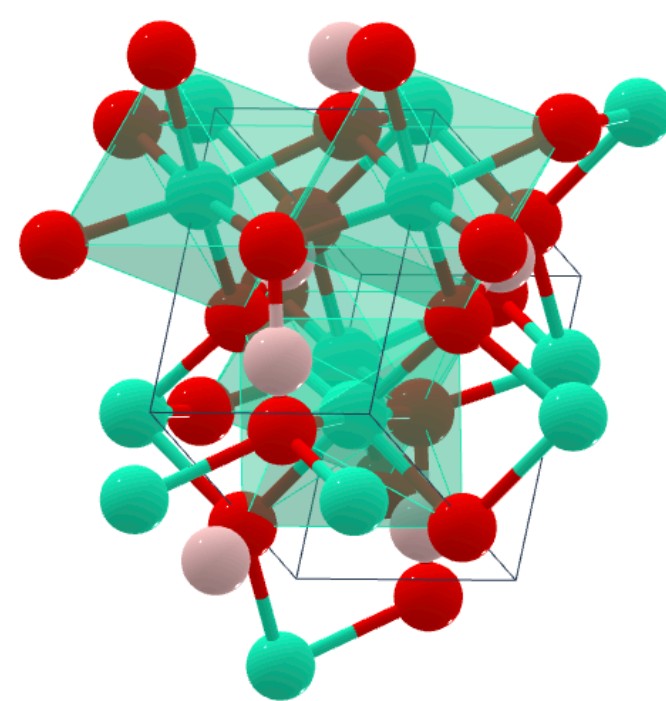
Layers can cover up to $m_a \sim 10^{-5}$ eV, but many other approaches possible

Optimizing Material

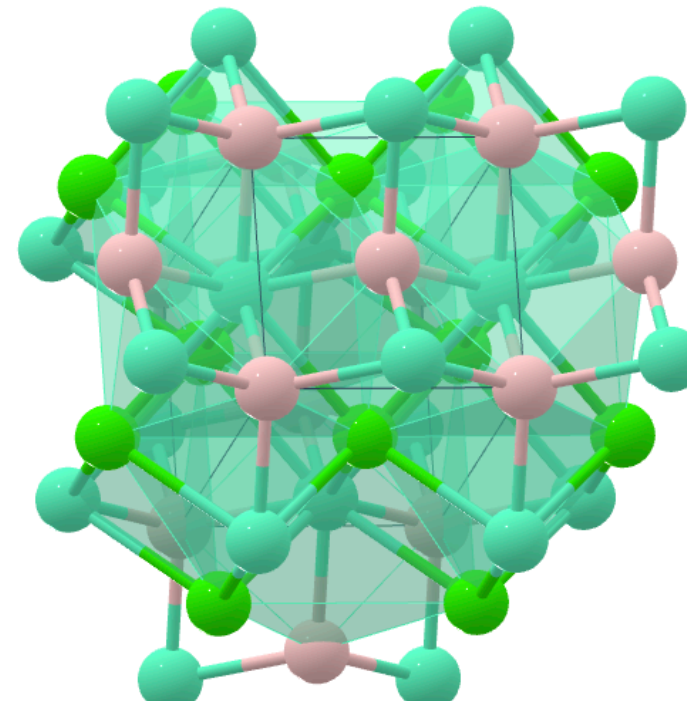
Material only has to be insulating; prefer high n_0 , low ϵ



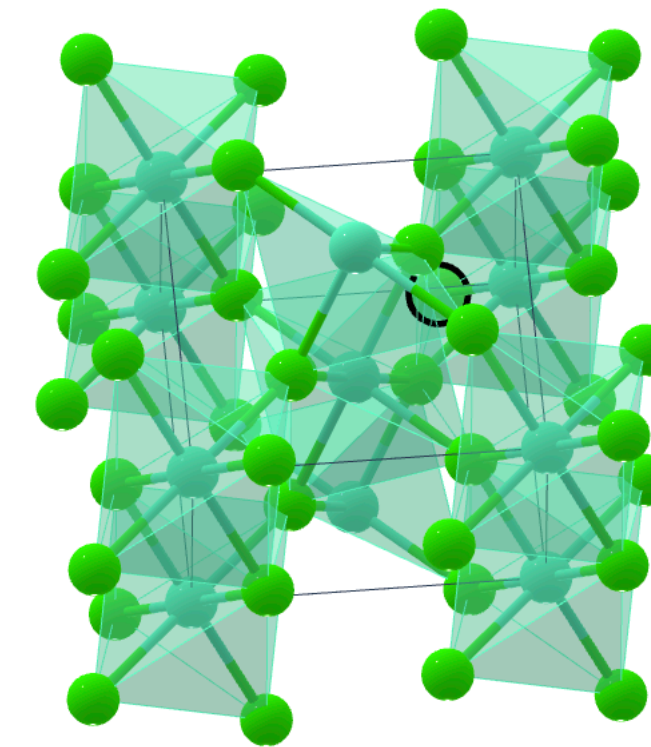
DyN



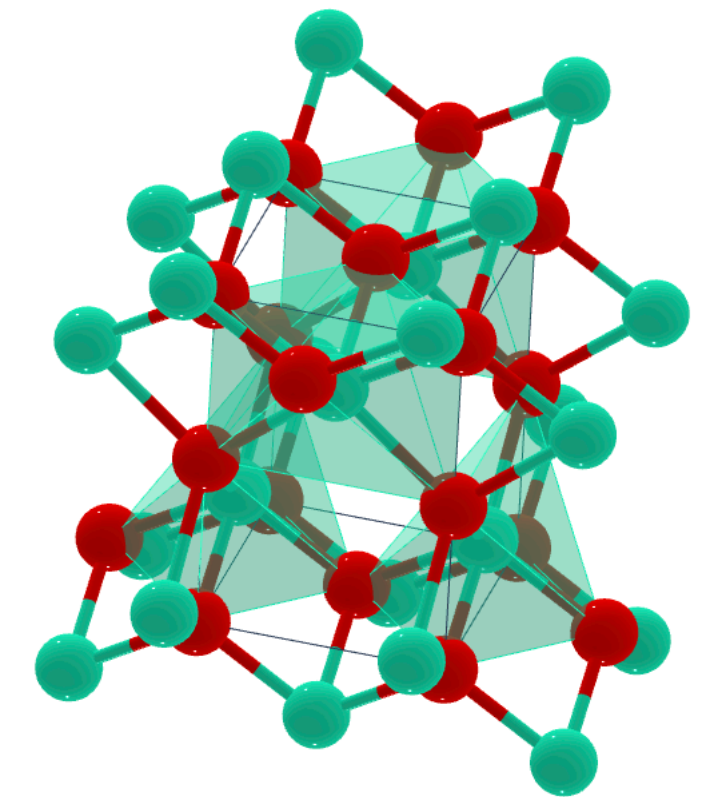
Dy₂O₃



DyHO₂



EuHCl



EuCl₂

These are simple, stable, and commercially available, but many alternatives exist

Dielectric Loss

- Low dielectric loss tangents $\tan \delta \lesssim 1/Q$ required
- Intrinsic losses (phonons) fall rapidly as T decreases, depends on symmetry
- Extrinsic losses (defects, impurities, TLS) dominate at low T , depend on quality
- Losses depend on T , ω , and applied field; no suitable existing measurements
- But loss tangents $\sim 10^{-9}$ observed for sapphire, $\sim 10^{-8}$ for other crystals

Spin Polarization

Thermal spin polarization depends on B/T , is $\sim 1\%$ at typical haloscope conditions

Like other approaches, achieving good sensitivity absolutely requires order-one f_p

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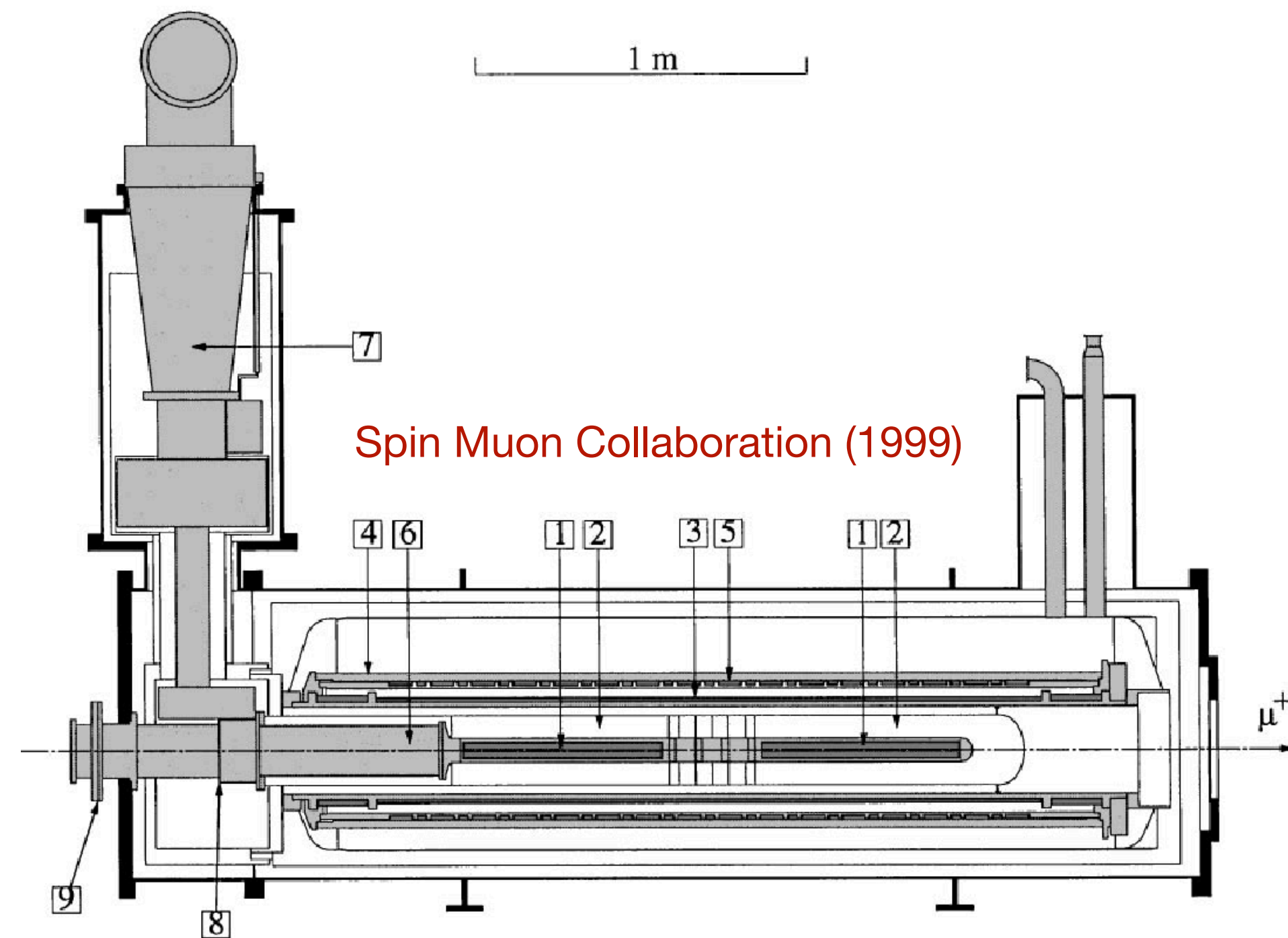
Frozen spin dynamic nuclear polarization

Polarize electron spins, transfer to nuclei with microwave radiation, and “freeze” result by lowering T

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More elaborate instrumentation, but meter-scale targets realized at CERN

Sensitivity Estimate

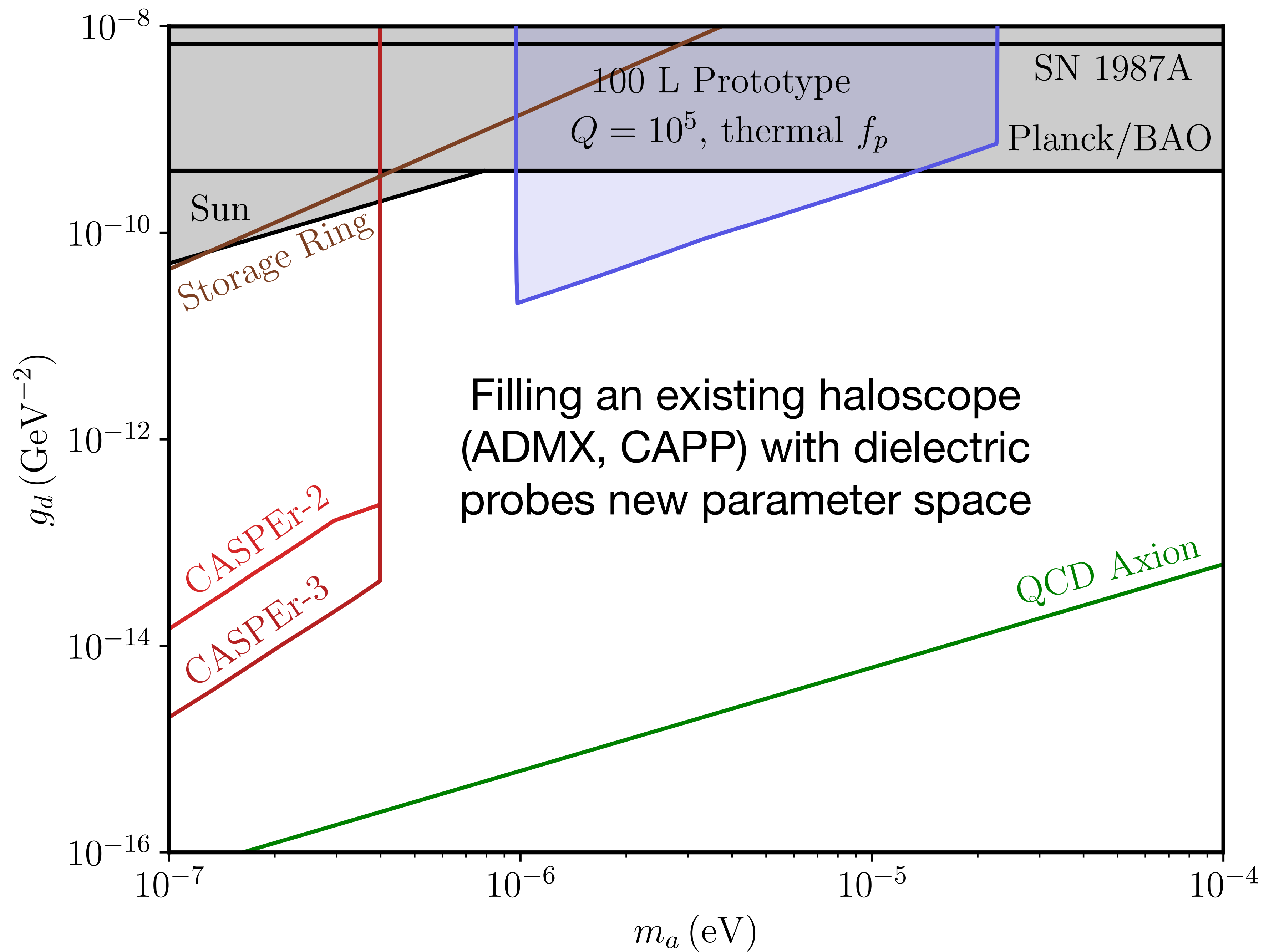
All noise sources besides thermal and amplifier noise vastly subdominant:

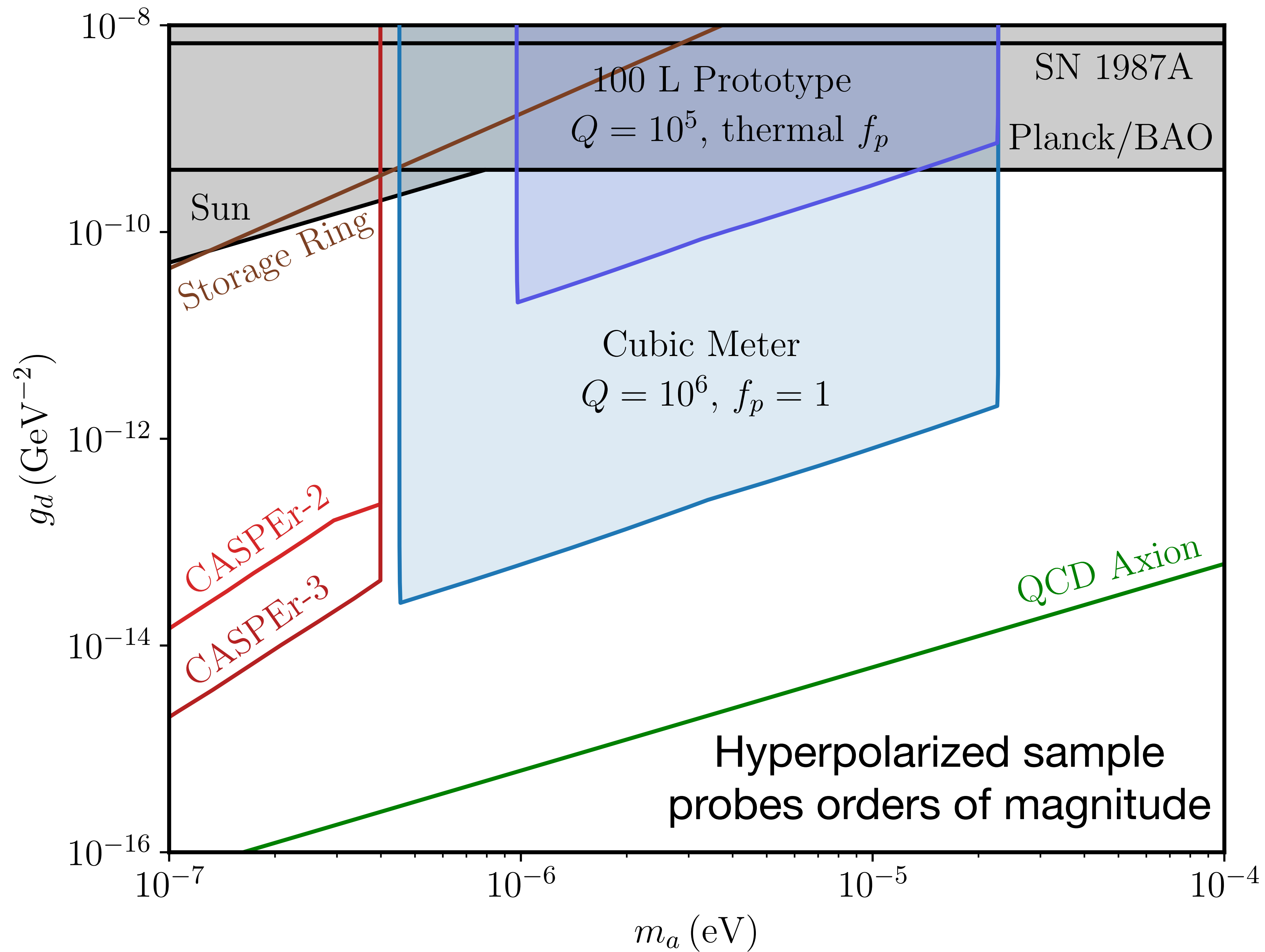
$$\text{SNR} \simeq \frac{P_{\text{sig}}}{T_n} \sqrt{\frac{t_{\text{int}}}{\Delta\omega_a}}, \quad T_n = T + T_{\text{amp}}$$

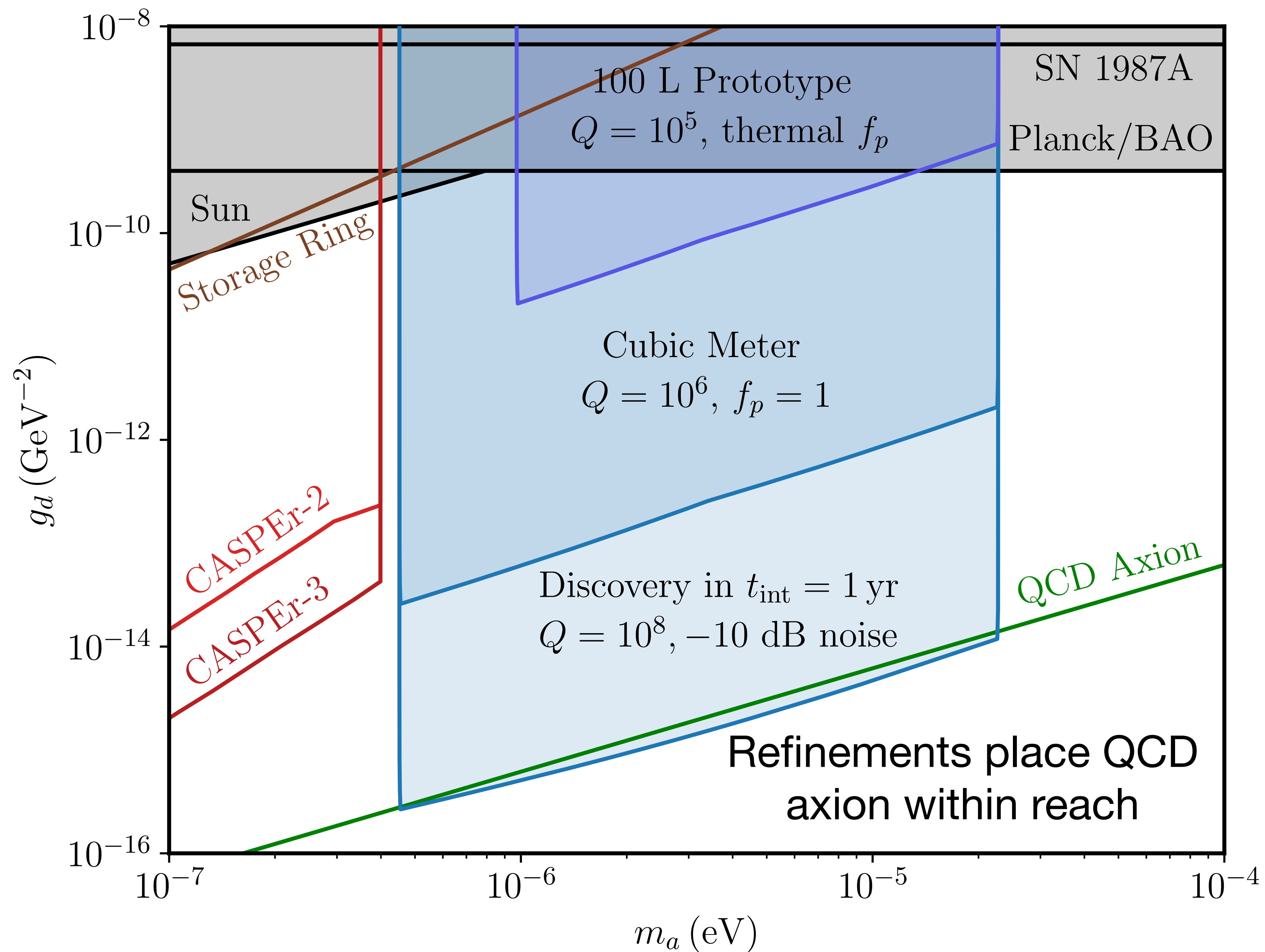
Assume quantum-limited amplifier, $T_{\text{amp}} = m_a$, and usual scanning procedure

$$\Delta\omega_s = \frac{m_a}{\max(Q, Q_a)}, \quad t_{\text{int}} = \frac{t_e}{\min(Q, Q_a)}$$

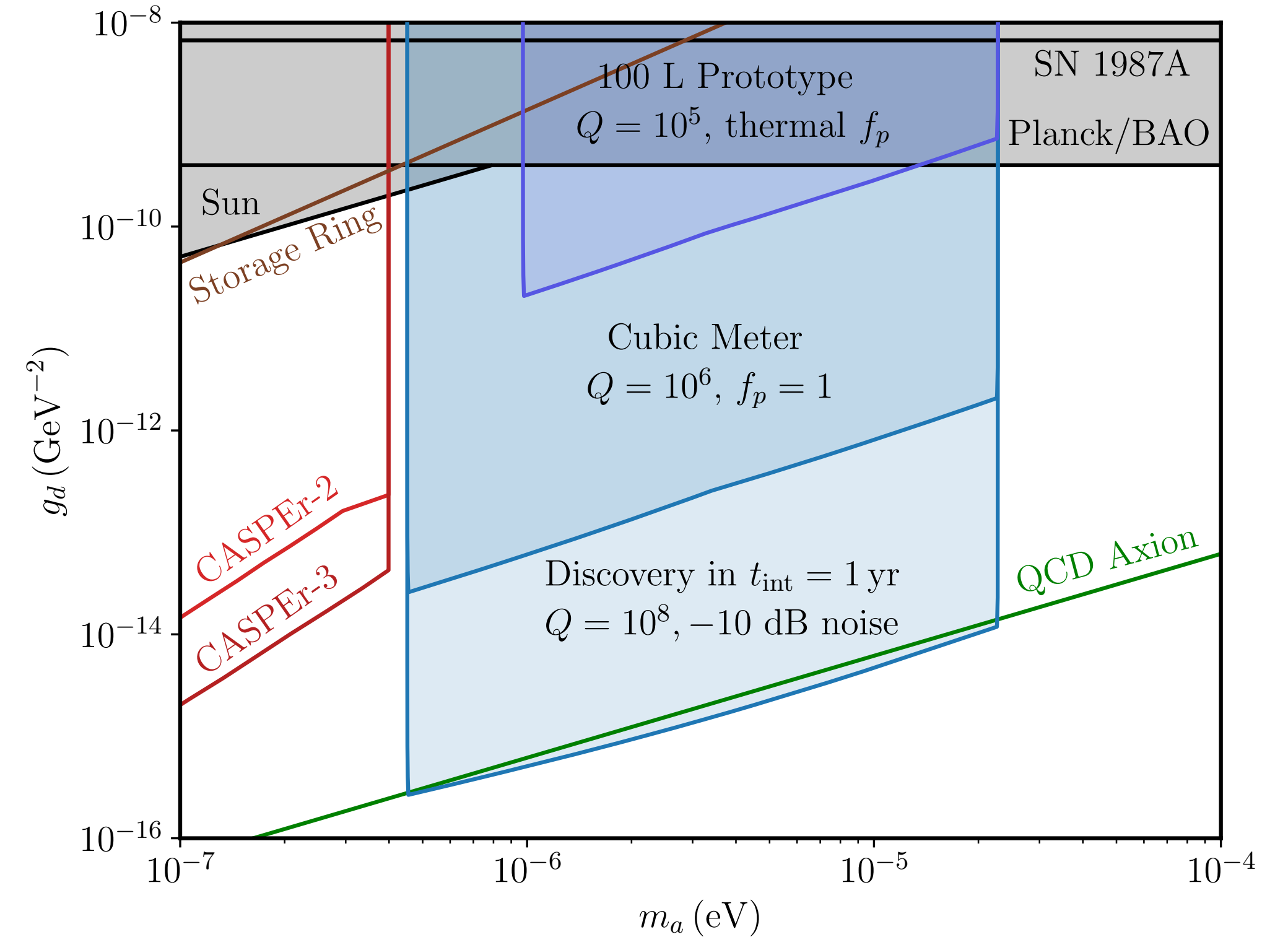
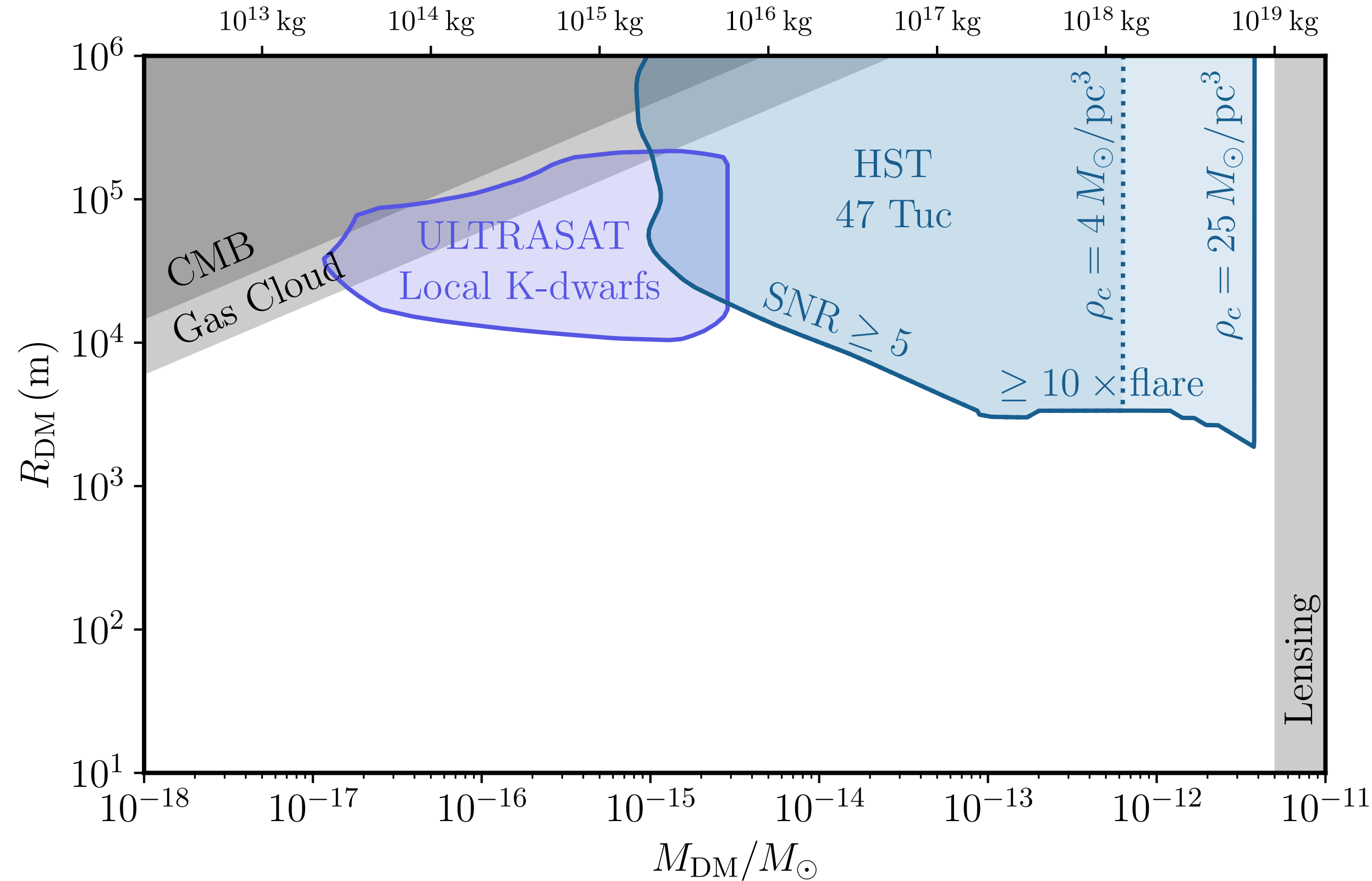
which implies $\text{SNR} \propto \sqrt{Q Q_a t_e}$ (overcoupling enhances by $\sqrt{T/m_a}$)







Conclusion



Exciting opportunities from the union of particle physics with astrophysics, hydrodynamics, nuclear physics, atomic physics, instrumentation... and much more!

Backup Slides

Schiff Moments

- Schiff moment defined by

$$\mathbf{S} = \frac{1}{10} \int d^3\mathbf{x} \rho_N(\mathbf{x}) r^2 \left(\mathbf{x} - \frac{5}{3} \frac{\mathbf{d}_N}{Ze} \right)$$

- Leads to atomic EDM via

$$V_S = - \sum_{i=1}^Z e \mathbf{S} \cdot \nabla \delta^3(\mathbf{x}_i), \quad D = - e \sum_{i=1}^Z e \mathbf{x}_i$$

$$d_A \sim \sum_n \frac{\langle n | V_s | 0 \rangle \langle 0 | D | n \rangle}{E_n - E_0} \propto Z^2 S, \quad d_A \simeq - (0.27 \times 10^{-3} e \text{ fm}) \frac{\langle S_z \rangle}{e \text{ fm}^3}$$

Schiff Moments

- Schiff moments induced by axion-mediated $\hat{P}$, $\hat{T}$ violating internucleon interaction

$$\langle S_z \rangle \sim \sum_n \frac{\langle n | V_a | 0 \rangle \langle 0 | S_z | n \rangle}{E_n - E_0}$$

- For spherical nuclei, parametric estimate is

$$\langle n | V_a | 0 \rangle \sim \frac{10^{-2} \theta_a}{m_n R_0} \frac{A}{m_\pi^2 R_0^3}, \quad \langle 0 | S_z | n \rangle \sim e R_0^3, \quad E_n - E_0 \sim \frac{A}{m_\pi^2 R_0^3}$$

$$\langle S_z \rangle \sim 10^{-2} \frac{e R_0^2}{m_n} \theta_a$$

Schiff Moments

- Much higher results for octupole-deformed nuclei with intrinsic Schiff moments,

$$\langle S_z \rangle = S_{\text{int}} \langle n_z \rangle, \quad S_{\text{int}} \propto ZeR_0^3, \quad \langle n_z \rangle \sim \sum_n \frac{\langle n | V_a | 0 \rangle \langle 0 | n_z | n \rangle}{E_n - E_0}$$

- Various numeric factors cancel, leaving Z-enhanced result

$$\langle S_z \rangle \sim 10^{-2} \frac{ZeR_0^2}{m_n} \theta_a, \quad |d_A| \sim (\text{few} \times 10^{-3}) \text{ e fm} \times \theta_a \left(\frac{Z}{10^2} \right)^3 \left(\frac{A}{10^2} \right)^{2/3}$$

- Comparable to d_n , possibly larger for very heavy nuclei